

NASH BARGAINING PREFERENCE OPTIMIZATION

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ABSTRACT

Aligning a large language model (LLM) with multiple objectives requires choosing a compromise while accounting for gains and losses on each objective. General pairwise preferences need not admit a scalar ranking, and evaluation against a fixed reference can miss weaknesses exposed by other responses. We introduce Nash Bargaining Preference Optimization (NBPO), which evaluates each objective against an opponent that adapts to the policy and maximizes the product of improvements over a shared reference policy. The policy and its opponents depend on one another; we directly optimize the probabilities of sampled responses, then train the LLM to reproduce their relative probability changes. Under the stated assumptions, the population optimum improves each objective’s game value over the reference and is Pareto efficient in these values. Exact population updates with a constant step size achieve an $O(1/T)$ bargaining-objective gap. Controlled experiments support adaptive-game representations as nontransitivity increases. **TO-DO:** more experiment results.

GPT: R01 — Review roadmap and central claim. The strongest possible story is not simply “Nash bargaining has desirable axioms.” It is that a specific reference-surplus bargaining problem over general preference games can be solved reliably enough to improve a deployed LLM. The present evidence leaves two links unresolved: whether the practical training procedure realizes the proposed update, and whether adaptive game values help beyond fixed-reference Nash. Organize the paper around a falsifiable sequence: demonstrate residual within-objective preference misspecification; define the particular compromise being chosen; solve the finite game accurately; measure neural realization error; then evaluate independent trade-offs. Prioritize these links before expanding benchmark coverage. The comments below address substantive framing, correctness, and experimental design; intentionally unfinished tables and routine placeholders are not the focus.

Response: R01 — Partially addressed. The main experiments now separate generated-policy quality, bargaining trade-offs, and adaptive-opponent mechanisms. UF is explicitly a score-induced-supervision comparison; controlled nontransitivity and the independent cross-play template are central. Natural-policy mechanism and external-baseline outcomes remain unmeasured, so the structure does not imply those claims are resolved.

1 INTRODUCTION

GPT: R02 — Distinguish model misspecification from noisy labels. The BT criticism in the next paragraph needs a probabilistic qualification. A BT model can assign nonzero probability to every observed binary outcome, including a sampled three-edge cycle; it cannot represent a population preference matrix whose strict majority relation cycles. Thus one contradictory annotation is not evidence that scalar reward modeling is misspecified. State the restriction at the level of P_k , not the requirement to satisfy every sampled label. Later compare a fitted BT model with held-out pair probabilities or repeated, order-balanced judgments. This distinction prevents a reviewer from dismissing the motivation as ordinary label noise, and makes clear what the game-based representation is intended to preserve.

Response: R02 — Partially addressed. The introduction distinguishes population misspecification from sampled label noise and identifies UF as a transitive-supervision comparison. The separate 60,000-judgment audit resolves neither repeated-cycle excess nor a GPM predictive advantage. Acyclic constructed labels are not evidence that fresh preferences on the same prompts must be transitive.

Reinforcement Learning from Human Feedback (RLHF) commonly learns scalar rewards from pairwise comparisons (Christiano et al., 2017; Ouyang et al., 2022). Direct Preference Optimization removes the separate reward model but retains a Bradley–Terry (BT) preference model (Rafailov et al., 2023; Bradley & Terry, 1952). For each prompt and objective, BT assigns one scalar score to each response and models preference probabilities through score differences. It permits noisy, contradictory individual labels, but its population preferences cannot have a cyclic strict majority relation.

Consider the prompt “How can I learn Python loops?” and three correct responses: A gives a short worked example, B a detailed conceptual explanation, and C a guided practice exercise. All annotators use the same overall-helpfulness rubric but differ in how they prefer to learn. Suppose three equally sized groups have fixed rankings: those seeking an immediate example prefer A, then B, then C; those seeking conceptual understanding prefer B, then C, then A; and those preferring practice prefer C, then A, then B. Two thirds therefore prefer A to B, two thirds B to C, and two thirds C to A. Every group’s ranking is transitive, yet the population majority cycles. This hypothetical construction illustrates how annotator heterogeneity within one rubric can produce probabilities that no shared BT reward can represent. PROSPER reports that LLM-judge comparisons on WildChecklists can remain cyclic even when each checklist item is evaluated separately (Zhang et al., 2026).

Nash Learning from Human Feedback (NLHF) retains general pairwise payoffs, and INPO, SPPO, and related methods provide scalable equilibrium-seeking updates (Munos et al., 2024; Zhang et al., 2025; Wu et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Multiple objectives additionally require a compromise rule. Existing methods use scalarization, constraints, policy interpolation, preference conditioning, or Blackwell criteria (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024; Agnihotri et al., 2026; Zhang et al., 2026).

GPT: R03 — The opening cycle currently changes the criterion between edges. Relevance justifies $A \succ B$, completeness justifies $B \succ C$, and safety justifies $C \succ A$. This establishes conflict across attributes, but not necessarily a cycle within one of the separate objective oracles used by NBPO. Specify one fixed rubric under which all three comparisons occur, or explicitly separate attribute aggregation from residual within-rubric intransitivity. PROSPER already studies this distinction empirically (Zhang et al., 2026). For our data, audit complete triples under the actual training rubric and compare reproducible majority cycles with an appropriate BT/noise control. An illustrative example is useful, but the paper needs evidence that the representation problem remains after splitting feedback into the proposed objectives.

Response: R03 — Partially addressed. The hypothetical example uses one helpfulness rubric. UF is now explicitly a score-induced alignment control, separate from the direct-feedback audit and controlled cyclic games. The audit finds no significant repeated-cycle excess or order-robust witness; residual cyclicity in natural training feedback remains unestablished.

We use classical Nash bargaining to maximize the product of improvements over a common fallback (Nash, 1950). The bargaining parties are objective rubrics, with annotators supplying comparisons for each rubric. We give these objectives equal bargaining weight to study a compromise without

externally specified priorities. Equal weight is a modeling choice: duplicating or splitting a rubric changes its influence. Prior work studies disagreement-adjusted Nash welfare over learned party rewards and bargaining over task-gradient improvements (Zhong et al., 2026; Navon et al., 2022).

GPT: R04 — Equal bargaining power is a choice, not absence of priorities. Prefer “does not require user-specified scalarization weights” or “equal bargaining weight for the specified objectives” to “priority-free.” Duplicating objective 1 changes its contribution from $\log s_1$ to $2 \log s_1$, generally changing the policy; splitting one rubric into several correlated rubrics has a similar effect. Specify whether the bargaining parties represent objectives, judges, or stakeholder groups, and justify equal treatment at that level. A targeted sensitivity test should merge or duplicate an overlapping rubric and vary the disagreement policy. When varying the reference, distinguish its role as the fallback from its role as the opponent anchor. These are substantive design choices that affine invariance does not remove.

Response: R04 — Partially addressed. The introduction identifies objective rubrics as bargaining parties, states that equal weight is a modeling choice, and separates the reference’s disagreement and opponent-anchor roles. Rubric duplication/merging and separate reference-sensitivity experiments are explicitly pending in Appendix A.3.

Nash Bargaining Preference Optimization (NBPO) bargains over reference-relative *policy game values*. Each objective evaluates the learner against an adaptive opponent that emphasizes responses exposing its weaknesses, while a KL penalty keeps the opponent near the reference policy. The reference thus serves both as the fallback for measuring improvement and as the anchor restricting the opponent. These roles can be varied separately (Appendix A.3). Improvement concerns this adaptive game evaluation, not necessarily direct win rate against the reference; population guarantees do not imply that every prompt improves.

INPO eliminates the policy normalizer through pairwise log-ratios (Zhang et al., 2025), and PROSPER combines KL-regularized adversarial comparisons, exponential reweighting, and regression-based policy fitting (Zhang et al., 2026). NBPO adds a logarithmic bargaining objective that is undefined at the unchanged reference and couples each policy update to its adaptive opponents. We derive a constrained proximal formulation and its dual, yielding objective weights shared across prompts. At fixed weights, we directly optimize sampled-response probabilities while accounting for the opponents’ dependence on them, then fit the LLM after freezing the target. This replaces an undamped policy–opponent alternation that can be unstable even when its fixed point is optimal. Numerical solution accuracy, neural fitting error, and independent policy quality require separate checks.

Our contributions are:

GPT: R05 — Attribute inherited ingredients and isolate the new technical increment. INPO already eliminates an intractable normalizer through pairwise log-ratios (Zhang et al., 2025); PROSPER already combines KL-regularized adversarial comparisons, exponential reweighting, and regression-based policy fitting (Zhang et al., 2026). Therefore, presenting those ingredients individually as the main contribution invites an incremental-combination objection. Explain precisely what the nonlinear reference-surplus objective adds: the zero-surplus boundary and the coupled proximal subproblem that must be solved despite policy-dependent opponents. Identify which derivation addresses this additional difficulty and which computation implements it. The current exact theorem and conditional fixed-point argument do not by themselves establish that the stated practical alternation solves this subproblem; R11–R12, R14, and R16 explain the missing link.

Response: R05 — Addressed. The introduction now attributes pairwise log-ratio fitting to INPO and the adversarial reweighting ingredients to PROSPER. The stated increment is the reference-surplus log objective and its coupled proximal dual, implemented through direct finite-pool optimization with a frozen neural target; numerical and neural errors are kept separate.

- We formulate multi-objective alignment as Nash bargaining over objective-wise regularized preference-game values, separating within-objective preference representation from across-objective compromise selection.
- We derive the dual of the coupled bargaining update and implement its inner solve as small strictly concave optimization problems over sampled responses. The resulting target is fitted through pairwise log-ratio regression, with numerical accuracy distinguished from neural realization error.

- For exact population updates with a constant step size, we prove that the gap in the bargaining objective at the final iterate decreases inversely with the number of updates. Controlled experiments examine adaptive game values as nontransitivity increases, while UltraFeedback and human-labeled SAFERLHF experiments evaluate neural realization and objective-wise policy outcomes.

2 BACKGROUND

General preference models. We use contextual-bandit notation. A prompt (context) x is drawn from a distribution $\mathcal{D}$, a response y is an action from a finite set $\mathcal{Y}_x$, and $\pi(y | x)$ is the policy. The reference policy is denoted by π_{ref} . For the finite-dimensional population analysis, the support of $\mathcal{D}$ and every $\mathcal{Y}_x$ are finite, and

$$\Pi = \prod_{x \in \text{supp}(\mathcal{D})} \Delta(\mathcal{Y}_x)$$

is the contextual policy simplex. Objective $k \in [K]$ is represented by a pairwise preference model $P_k(y \succ z | x) \in [0, 1]$ satisfying

$$P_k(y \succ z | x) + P_k(z \succ y | x) = 1, \quad P_k(y \succ y | x) = \frac{1}{2}. \quad (1)$$

We center each comparison at a tie,

$$\Delta_k(y, z | x) = P_k(y \succ z | x) - \frac{1}{2}, \quad \Delta_k(y, z | x) = -\Delta_k(z, y | x), \quad (2)$$

and define the expected objective- k payoff of learner policy π against comparator policy ν as

$$J_k(\pi, \nu) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot | x), z \sim \nu(\cdot | x)} \Delta_k(y, z | x). \quad (3)$$

Thus $J_k(\pi, \nu) > 0$ means that π is preferred to ν on average under objective k . This representation does not assume a scalar reward or transitive preferences.

Bargaining problems. A K -dimensional bargaining problem consists of a feasible value set $\mathcal{G} \subseteq \mathbb{R}^K$ and a disagreement point $d \in \mathcal{G}$, the outcome obtained if no agreement is reached. For a feasible value vector v , define its objective-wise surplus over disagreement as

$$s_k(v) := v_k - d_k, \quad s(v) = (s_1(v), \dots, s_K(v)). \quad (4)$$

A value vector is individually rational when $s(v) \geq 0$ componentwise, and strictly individually rational when $s(v) > 0$. On a compact, convex feasible set that is downward-closed within $\{v : v \geq d\}$ and contains a strictly individually rational point, the classical Nash solution is

$$v^N \in \arg \max_{v \in \mathcal{G} : s(v) > 0} \sum_{k=1}^K \log s_k(v). \quad (5)$$

Equivalently, it maximizes the product of the objective-wise surpluses (Nash, 1950).

3 OBJECTIVE-WISE PREFERENCE GAMES

We first convert each objective’s pairwise preference model into a scalar policy value without imposing a scalar reward on individual responses. For objective k , an adaptive comparator ν searches for responses against which the learner performs poorly. The reference policy π_{ref} anchors this comparator and later serves as the fallback in the bargaining problem.

Definition 1 (Entropy-regularized game value). *For objective k and temperature $\beta_k > 0$, define*

$$V_{k, \beta_k}(\pi) = \min_{\nu} \{J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x))\}. \quad (6)$$

The comparator minimizes the learner’s pairwise payoff, so it emphasizes responses that expose weaknesses of the current policy. The KL term prevents it from moving arbitrarily far from π_{ref} . Accordingly, a large β_k produces a local evaluation near the reference, whereas $\beta_k \downarrow 0$ approaches the unregularized worst-comparator value.

For a candidate comparator response z , let

$$m_k^\pi(z \mid x) := \mathbb{E}_{y \sim \pi(\cdot \mid x)} \Delta_k(y, z \mid x) \quad (7)$$

be the learner’s expected margin against z . Smaller values identify stronger challenges. The inner problem in (6) has the closed-form solution

$$\nu_{k,\pi}^*(z \mid x) = \frac{\pi_{\text{ref}}(z \mid x) \exp\{-m_k^\pi(z \mid x)/\beta_k\}}{\mathbb{E}_{z' \sim \pi_{\text{ref}}(\cdot \mid x)} \exp\{-m_k^\pi(z' \mid x)/\beta_k\}}, \quad (8)$$

$$V_{k,\beta_k}(\pi) = -\beta_k \mathbb{E}_{x \sim \mathcal{D}} \log \mathbb{E}_{z \sim \pi_{\text{ref}}(\cdot \mid x)} \exp\{-m_k^\pi(z \mid x)/\beta_k\}. \quad (9)$$

Thus the opponent is obtained by reweighting reference responses toward those with smaller margins, and the game value is a soft minimum of these margins.

Proposition 1 (Game-value structure). *For every k and $\beta_k \geq 0$, V_{k,β_k} is continuous and concave in π . For $\beta_k > 0$, its policy-gradient coordinate is*

$$[\nabla_\pi V_{k,\beta_k}(\pi)]_{x,y} = \mathcal{D}(x) Q_k^\pi(y \mid x), \quad Q_k^\pi(y \mid x) := \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot \mid x)} \Delta_k(y, z \mid x). \quad (10)$$

The proof is in Appendix A.1. The response-level score $Q_k^\pi(y \mid x)$ is the expected margin of response y against the current objective- k opponent; it therefore measures the local effect of increasing $\pi(y \mid x)$. An adaptive opponent can distinguish policies that have the same win rate against one fixed reference but different vulnerabilities to other responses. Appendix A.2 gives a finite cyclic example at the unregularized endpoint $\beta = 0$; this illustration is not used in the later assumptions or proofs.

4 NASH BARGAINING OVER GAME-VALUE IMPROVEMENTS

We now combine the objective-wise game values using the reference policy as the common fallback. For each objective,

$$d_k := V_{k,\beta_k}(\pi_{\text{ref}}), \quad s_k(\pi) := V_{k,\beta_k}(\pi) - d_k. \quad (11)$$

The surplus $s_k(\pi)$ is the improvement of π over the deployed reference under objective k .

GPT: R06 — Game-value improvement is not necessarily a direct win over the fallback. Qualify this sentence and the corresponding introduction claim: the improvement is in V_{k,β_k} , not necessarily in $J_k(\pi, \pi_{\text{ref}})$. A concrete one-prompt example uses the cyclic margin matrix

$$A = \begin{pmatrix} 0 & .5 & -.5 \\ -.5 & 0 & .5 \\ .5 & -.5 & 0 \end{pmatrix}, \quad q = (.8, .1, .1), \quad p = (.25, .43, .32), \quad \beta = .25.$$

Using q as the reference gives $V(q) \simeq -.0517901$ and $V(p) \simeq -.0417293$, hence positive surplus $s(p) \simeq .0100608$, but $J(p, q) = -.0385$: the direct reference win rate is only .4615. This is a full-support, finite-temperature example under the same judge. Explain early why robustness-adjusted value is the intended notion of improvement, and evaluate direct wins separately. The limitation is more basic than the absence of per-prompt safety guarantees.

Response: R06 — Addressed. The introduction explicitly limits improvement to adaptive game values and separates direct reference wins. Appendix A.2 now includes the full-support finite-temperature counterexample, with positive surplus but direct win probability 0.4615.

Assumption 1 (Joint improvement and support). *The reference π_{ref} has full support, and there exists a policy $\bar{\pi}$ such that $s_k(\bar{\pi}) > 0$ for every k .*

GPT: R07 — Provide a feasibility diagnostic and a defined failure outcome. Joint strict improvement is a substantive assumption, particularly for conflicting objectives or an already strong reference. A constrained formulation makes the exact log problem well-defined when a feasible improvement exists; it does not discover one automatically or make an infeasible problem feasible. On each finite response pool, solve a phase-I problem $\max_{\pi, \tau} \tau$ subject to $s_k(\pi) \geq \tau$ for every k . Report its margin, then distinguish a genuinely nonpositive optimum from numerical or neural failure to realize a positive one. Include an intentionally conflicting panel with no strict common improvement and specify the returned status/policy. At the reference boundary, report rejection/stalling frequency. Floors and clipped inverse-surplus weights cannot create feasibility.

Response: R07 — Partially addressed. Appendix A.5 defines a finite-pool maximum-margin feasibility audit, an analytic conflicting example, distinct failure statuses, and retention of the previous policy on uncertified output. Measured feasibility margins and boundary rejection/stalling rates still require execution artifacts; the constrained formulation is not claimed to create feasibility.

This is a substantive feasibility assumption. A finite-pool audit and failure outcomes are specified in Appendix A.5; the constrained formulation cannot make conflicting objectives jointly improvable. Under this condition, the NBPO population target is

$$\pi^N \in \arg \max_{\pi \in \Pi_+} F(\pi), \quad F(\pi) := \sum_{k=1}^K \log s_k(\pi), \quad \Pi_+ := \{\pi \in \Pi : s_k(\pi) > 0, \forall k\}. \quad (12)$$

Equivalently, introducing auxiliary lower bounds $u_k > 0$ yields

$$(\pi^N, u^N) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \sum_{k=1}^K \log u_k \quad \text{s.t.} \quad u_k \leq s_k(\pi), \quad k \in [K], \quad \pi \in \Pi. \quad (13)$$

Because the objective is increasing in every u_k , each constraint is tight at an optimum and $u_k^N = s_k(\pi^N)$.

To connect (12) to the classical bargaining problem, define

$$V_\beta(\pi) = (V_{1,\beta_1}(\pi), \dots, V_{K,\beta_K}(\pi)), \quad d = (d_1, \dots, d_K),$$

and the downward-closed feasible value set

$$\mathcal{G}_\beta := \{v \in \mathbb{R}^K : d \leq v \leq V_\beta(\pi) \text{ componentwise for some } \pi \in \Pi\}. \quad (14)$$

Proposition 2 (Policy-induced Nash bargaining). *Under Assumption 1, $\mathcal{G}_\beta$ is compact, convex, downward closed above d , and nondegenerate. The policies solving (12) realize the classical Nash solution of $(\mathcal{G}_\beta, d)$. Consequently, every optimizer has positive surplus in every objective and is Pareto optimal in game-value space; all optimizers induce the same game-value vector; and the optimizer set is unchanged by independent transformations*

$$(V_{k,\beta_k}, d_k) \mapsto (c_k V_{k,\beta_k} + b_k, c_k d_k + b_k), \quad c_k > 0.$$

Appendix A.3 verifies the feasible-set conditions. Once those conditions hold, the stated outcome properties are standard consequences of the classical Nash bargaining solution (Nash, 1950); we do not claim them as new bargaining axioms. These properties concern population game values. Scaling primitive margins also requires scaling β_k ; arbitrary recalibration of judge probabilities is not covered (Appendix A.3). Appendix A.4 compares these guarantees with utilitarian and maxmin aggregation.

5 NASH BARGAINING PREFERENCE OPTIMIZATION

GPT: R08 — State exactly what the affine-invariance proposition covers. The proposition correctly transforms completed utility functions and disagreement values together. It does not imply invariance to arbitrary judge calibration changes. At the primitive game level, if $\Delta'_k = c_k \Delta_k$, one must also set $\beta'_k = c_k \beta_k$ to preserve the opponent and obtain $V'_k = c_k V_k$. With temperature fixed, instead $V_{c\Delta,\beta} = c V_{\Delta,\beta/c}$, which changes the game. Also, scaling a centered preference margin beyond its probability range is a change of payoff units, not necessarily a realizable new probability oracle. Keep the algebraic guarantee, but explain this distinction before connecting it to judge-specific units or the rescaling experiment.

Response: R08 — Addressed. The proposition is explicitly limited to joint transformations of completed utilities and disagreement values. Appendix A.3 now gives the required temperature scaling, the fixed-temperature identity, and the distinction between payoff-unit changes and valid probability oracles; arbitrary judge recalibration is excluded.

Optimizing (12) presents three practical difficulties. First, the reference has $s_k(\pi_{\text{ref}}) = 0$, so the logarithmic objective is undefined at initialization. Second, every game value contains an opponent that depends on the learner. Third, the population policy update must be realized by an autoregressive model. We handle these issues with a constrained proximal problem, a prompt-separable finite-pool inner solve, and neural log-ratio projection.

5.1 CONSTRAINED PROXIMAL UPDATE AND ADAPTIVE OBJECTIVE WEIGHTS

For $\pi \in \Pi_+$, Proposition 1 and the chain rule give

$$[\nabla_\pi F(\pi)]_{x,y} = \mathcal{D}(x) \sum_{k=1}^K \frac{Q_k^\pi(y | x)}{s_k(\pi)}. \quad (15)$$

The inverse-surplus factors emphasize objectives that have improved less. Because this expression cannot be evaluated at π_{ref} , outer stage t instead solves

$$\begin{aligned} (\pi_{t+1}, u_{t+1}) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \quad & \sum_{k=1}^K \log u_k - \frac{1}{\eta_t} D(\pi \| \pi_t) \\ \text{s.t.} \quad & u_k \leq s_k(\pi), \quad k \in [K], \\ & \pi \in \Pi, \end{aligned} \quad (16)$$

where $\pi_0 = \pi_{\text{ref}}$, $\eta_t > 0$, and $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$.

The dual is

$$\min_{\lambda \in \mathbb{R}_{++}^K} \phi_t(\lambda), \quad \phi_t(\lambda) = - \sum_{k=1}^K \log \lambda_k - K + \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}, \quad (17)$$

where $\lambda = (\lambda_1, \dots, \lambda_K) \in \mathbb{R}_{++}^K$ is the vector of Lagrange multipliers. For fixed λ , let

$$\pi_t^*(\lambda) := \arg \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (18)$$

The KL term makes the maximizing policy $\pi_t^*(\lambda)$ unique. Danskin’s theorem (Danskin, 1967) then allows us to differentiate $\phi_t(\lambda)$ while holding this maximizing policy fixed, yielding

$$\nabla_k \phi_t(\lambda) = s_k(\pi_t^*(\lambda)) - \frac{1}{\lambda_k}. \quad (19)$$

The complete dual derivation is in Appendix A.5.

Proposition 3 (Inverse-surplus objective weights). *At an optimal primal–dual solution of (16),*

$$u_{t+1,k} = s_k(\pi_{t+1}), \quad \lambda_{t+1,k} = \frac{1}{u_{t+1,k}} = \frac{1}{s_k(\pi_{t+1})}. \quad (20)$$

Thus the constrained problem recovers the inverse-surplus weighting in (15) without selecting objective weights in advance.

5.2 DIRECT FINITE-POOL INNER SOLVE AND NEURAL PROJECTION

GPT: R09 — Keep three distributions distinct throughout the implementation. The population proof optimizes over the full contextual simplex; a finite-pool solver optimizes categorical weights over sampled responses; the trained LLM defines a distribution over all possible responses. These are not interchangeable objects. Introduce separate notation for the exact inner solution, the finite-pool teacher, and the fitted neural policy. State which one generates each opponent and supplies each surplus in the dual update. This distinction is especially important with only four candidates and $R = 1$: a highly accurate teacher solve can coexist with a poor neural projection. It should be possible to trace one outer update from sampled data to teacher weights to neural parameters without inferring undocumented computations.

Response: R09 — Addressed. The method now separates the population inner solution $\pi_t^*(\lambda)$, finite-pool target $p_{t,x}^*(\lambda)$, and neural policy π_θ . Finite-pool opponents and prompt-averaged surpluses drive the numerical dual solve; one neural fit follows the frozen target, and its importance-reweighted pool distribution is diagnosed separately.

At outer stage t , draw learner occurrences $\mathcal{Y}_x = \{y_{x,i}\}_{i=1}^{n_x}$ from π_t and comparator occurrences $\mathcal{Z}_x = \{z_{x,j}\}_{j=1}^{m_x}$ from π_{ref} . Write $\hat{A}_{k,x}[i,j] = \hat{\Delta}_k(y_{x,i}, z_{x,j} | x)$. Retaining duplicate draws

with multiplicity gives empirical base measures $p_{t,x}(i) = 1/n_x$ and $\mu_x(j) = 1/m_x$. These are importance-sampling occurrence weights, not normalized LM likelihoods. For $p \in \Delta^{n_x}$, define

$$\hat{m}_{k,x}(p)_j := \sum_{i=1}^{n_x} p_i \hat{A}_{k,x}[i, j], \quad (21)$$

$$\hat{V}_{k,x}(p) := -\beta_k \log \sum_{j=1}^{m_x} \mu_{x,j} \exp\{-\hat{m}_{k,x}(p)_j / \beta_k\}. \quad (22)$$

The fixed disagreement estimate $\hat{d}_k$ comes from a reference–reference tensor, not an assumed zero. Let $\hat{\mathcal{D}}$ be the empirical prompt distribution and set $\hat{s}_k(p) = \mathbb{E}_{x \sim \hat{\mathcal{D}}} \hat{V}_{k,x}(p_x) - \hat{d}_k$. The shared multipliers aggregate these prompt averages; there is no separate Nash bargain at each prompt.

For fixed λ , the empirical counterpart of (18) separates across prompts:

$$p_{t,x}^*(\lambda) \in \arg \max_{p \in \Delta^{n_x}} \left\{ \sum_{k=1}^K \lambda_k \hat{V}_{k,x}(p) - \frac{1}{\eta_t} \text{KL}(p \| p_{t,x}) \right\}. \quad (23)$$

Each problem is strictly concave because of the KL term. The opponent is analytically eliminated inside $\hat{V}$, so direct optimization accounts for its dependence on p . We solve these small programs to numerical tolerance, then solve the shared dual condition $\hat{s}_k(p_t^*(\lambda)) = 1/\lambda_k$ in $\log \lambda$. A computational box $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$ requires reporting both unprojected stationarity and bound activity; projected stationarity alone does not certify the original problem. Appendix A.5 gives the residual checks and an explicit instability example for undamped policy–opponent alternation.

Once the dual converges, the finite-pool target defines the pairwise log-ratio change

$$\hat{h}_t^*(x, i, i') := \log \frac{p_{t,x}^*(i)}{p_{t,x}(i)} - \log \frac{p_{t,x}^*(i')}{p_{t,x}(i')}. \quad (24)$$

For a neural policy π_θ , let

$$h_t(\pi_\theta; x, y, y') := \log \frac{\pi_\theta(y | x)}{\pi_t(y | x)} - \log \frac{\pi_\theta(y' | x)}{\pi_t(y' | x)}. \quad (25)$$

GPT: R10 — Full-support identification does not follow from complete pairing within four candidates. All six pairs among four sampled responses determine their relative probabilities, but do not control mass on unseen responses. For example, if only pair (1, 2) is observed with zero log-ratio-change target under a uniform three-response anchor, every policy $(a, a, 1 - 2a)$ fits that pair while assigning different mass to response 3. State the support on which the regression-identification claim applies. Test candidate-pool expansion and fresh generations outside the training pool, and compare the exact teacher distribution with its neural realization. Separately qualify unbiasedness: the binary target has the displayed conditional mean under an exact opponent; the sampled, exponentially reweighted approximation in Appendix [original-draft ref.: app:labeling] requires its own error analysis (R34).

Response: R10 — Partially addressed. Identification is restricted to connected sampled support, with an explicit unseen-mass counterexample and a separate neural pool diagnostic. Fresh-response UF-4 evaluation now complements target fitting, but controlled candidate-pool expansion and quantitative finite-sampling error remain unreported; generated-policy gains do not establish full-support identification.

We project the finite-pool target into the LLM with

$$\mathcal{L}_{\text{reg}}^{(t)}(\theta) = \mathbb{E}_{\substack{x \sim \hat{\mathcal{D}} \\ (i, i') \sim \xi_x}} \left[\left(h_t(\pi_\theta; x, y_{x,i}, y_{x,i'}) - \hat{h}_t^*(x, i, i') \right)^2 \right]. \quad (26)$$

GPT: R11 — Critical: the displayed regression objective is not the objective used in the proof. The square is outside the expectation, giving $(\mathbb{E}[\text{residual}])^2$, whereas conditional-mean regression requires $\mathbb{E}[(\text{residual})^2]$. These are mathematically different: large positive and negative residuals can cancel in the first expression; with symmetric ordered-pair sampling, that objective can even become uninformative. Move the square inside the expectation and verify the actual training loss, not only the typography. The appendix’s argument that the regression minimizer recovers the desired pairwise target depends on minimizing squared error for individual examples. This is a substantive correctness issue because the policy-identification claim is otherwise unsupported.

Response: R11 — Partially addressed. Equation (26) now places each example’s squared residual inside the expectation, and the diagnostics use the same mean-squared-error definition. This fixes the manuscript’s objective; checking the actual training loss still requires the implementation, which was not supplied with these attachments.

The target is frozen during fitting. Complete pair coverage identifies only the reweighted sampled support, not unseen responses. At the inner optimum,

$$\hat{h}_t^*(x, i, i') = \eta_t \sum_{k=1}^K \lambda_k \left(\hat{Q}_{k,x}^*(i) - \hat{Q}_{k,x}^*(i') \right), \quad (27)$$

where $\hat{Q}_{k,x}^* = \hat{A}_{k,x} \nu_{k,x}(p_{t,x}^*)$. This identity checks the solver target independently. Pairwise probabilities may come from direct human comparisons or a general pairwise preference model; NBPO never requires them to decompose into scalar response rewards.

GPT: R12 — Critical: fixed-point optimality does not establish convergence of the proposed alternation. The appendix shows that an exact fixed point satisfies the inner optimality conditions, not that recomputing best responses reaches it. The iteration can actually be unstable. Use the RPS matrix A in R06, with one objective, reference/outer anchor $q = (.34, .33, .33)$, $\beta = .25$, and $\eta\lambda = 4$. The fixed-opponent map is

$$T(p) = \text{softmax}(\log q + 4A \text{softmax}(\log q + Ap/\beta)).$$

Its fixed point is approximately $(.33618258, .32907171, .33474571)$, but both tangent Jacobian eigenvalues are approximately -1.33316 , implying local instability of undamped iteration. Full support holds, and this point improves value from about -3.30×10^{-5} at q to 1.42×10^{-5} , so the joint-improvement assumption does not exclude the example. Replace “produces a fixed point” with the conditional statement proved. Supply a justified inner solver, an appropriate damping/contraction condition, or a different convergent scheme, and validate its residuals. Merely increasing R is not a principled repair.

Response: R12 — Addressed. The unconditional alternation claim is replaced by direct optimization of the strictly concave finite-pool objective. Appendix A.5 includes the unstable fixed-point example, its approximately -1.333163 tangent eigenvalues, the objective’s gradient/Hessian, and explicit numerical-residual requirements.

Neural pool diagnostics use $\tilde{p}_{\theta,x}(i) \propto p_{t,x}(i)\pi_{\theta}(y_{x,i} \mid x)/\pi_t(y_{x,i} \mid x)$, not normalized raw LM likelihoods. Appendix E compares regression with weighted behavior cloning of the same target.

GPT: R13 — A bounded dual box can change the optimization problem. The exact multiplier satisfies $\lambda_k^* = 1/s_k(\pi_{t+1})$. Therefore a legitimate positive surplus below 10^{-3} can require a multiplier above the implemented upper bound 10^3 ; strict feasibility supplies no numerical lower bound on surplus. At a box boundary, projected stationarity need not imply the original condition $s_k - 1/\lambda_k = 0$. Report coordinatewise boundary-hit rates, unprojected KKT residuals, and sensitivity to the bounds. Explain whether the 5×10^{-3} surplus floor changes the objective, only stabilizes estimation, or is used only in the rescaling diagnostic. Justify the box from a measured feasible margin where possible. Do not use clipped-solver convergence as a certificate for the original Nash proximal problem.

Response: R13 — Partially addressed. Projected box stationarity is distinguished from the original unprojected KKT condition. The added UF-4 reconstruction reports multipliers 7.30–11.04, a 3.74×10^{-14} dual residual, and a 1.88×10^{-10} maximum inner stationarity spread. Bound-sensitivity and boundary-hit sweeps, and the historical surplus floor’s role, still require verification.

Acceptance is empirical, uses development rather than final-test data, and provides no population certificate. Appendix A.5 defines failure reporting. Historical runs require artifact verification of this execution contract; all reported LLM runs use $T = 1$.

Algorithm 1 Nash Bargaining Preference Optimization

Require: Reference π_{ref} , pairwise models $P_{1:K}$, temperatures $\beta_{1:K}$, outer stages T .

```
1: Set  $\pi_0 \leftarrow \pi_{\text{ref}}$ ; construct fixed reference pools and  $\hat{d}_{1:K}$ .
2: for  $t = 0, \dots, T - 1$  do
3:   Draw learner pools from  $\pi_t$ , construct cross tensors, initialize  $\lambda > 0$ .
4:   repeat
5:     Solve (23) independently for every prompt.
6:     Update shared  $\lambda$  using  $\hat{s}_k - 1/\lambda_k$ ; check residuals and bounds.
7:   until unprojected KKT tolerance is met or the numerical budget is exhausted
8:   On uncertified output, audit feasibility, retain  $\pi_t$ , and return its failure status.
9:   Construct  $\hat{h}_t^*$  by (24); verify (27).
10:  Fit one neural candidate  $\tilde{\pi}_{t+1}$  using (26).
11:  Measure post-fit diagnostics on development data; accept only if declared checks pass.
12:  Set  $\pi_{t+1} \leftarrow \tilde{\pi}_{t+1}$  if accepted, otherwise retain  $\pi_t$  and record rejection.
13: end for
14: return  $\pi_T$ .
```

GPT: R14 — Reconcile Algorithm 1 with the reported computation. The pseudocode calls POLICYFIT at every dual iteration, but the configuration lists 40,000–300,000 dual iterations and only 300 neural optimizer updates per stage. This suggests a finite-pool teacher solve followed by one neural fitting stage, rather than a fresh neural inner solve for every multiplier update. If that is the implementation, show it explicitly as a teacher solver plus a neural realization algorithm; if not, explain the actual update counts and costs. Recompute surplus, policy-opponent consistency, and KKT diagnostics after neural realization. A residual near 10^{-15} for the frozen teacher is not a certificate for the deployed LLM. Audit the implementation before deciding which algorithmic claim the paper can defend.

Response: R14 — Partially addressed. The algorithm explicitly separates the finite-pool solve from one subsequent neural fit. UF-4 now reports independently reconstructed teacher residuals and 1,250 neural updates, followed by development fitting diagnostics and fresh-response evaluation. Full post-fit surplus, opponent-consistency, and KKT checks remain pending; teacher residuals do not certify the neural policy.

GPT: R15 — Test the acceptance safeguard near the boundary. The existing disclaimer correctly avoids claiming a finite-sample guarantee, but positivity of noisy estimates is still the decisive practical acceptance criterion. With four comparators and small surpluses, an update can be falsely accepted or a genuinely feasible update repeatedly rejected. On known finite games, measure both error rates as functions of pool size, L , temperature, and true feasibility margin. Include all objectives jointly and report rejection frequency, especially at $t = 0$. Keep the monitoring data separate from final evaluation; repeatedly using a held-out set to choose policies makes it validation data. A conservative simultaneous lower-confidence-bound check is worth testing, but do not apply a bounded-mean formula blindly to biased nonlinear game-value estimates.

Response: R15 — Partially addressed. Acceptance is now explicitly empirical and uses development data, with numerical-unresolved and neural-rejected outcomes recorded separately. Appendix A.5 acknowledges nonlinear estimation bias, but the requested near-boundary false-acceptance/rejection and joint-objective error-rate sweeps remain unmeasured.

Last-iterate convergence. The following Bregman proximal-point guarantee concerns exact population solves, not finite-pool fitting or empirical acceptance; T counts exact solves rather than neural updates. Under Assumption 1, with $\pi_0 = \pi_{\text{ref}}$ and (16) solved exactly at every outer iteration with arbitrary $\eta_t > 0$, the Nash welfare is monotone, $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and $F(\pi^*) - F(\pi_T) \leq D(\pi^* || \pi_{\text{ref}}) / \sum_{t < T} \eta_t$: a constant proximal step gives an $O(1/T)$ final-iterate gap. Theorem 1 and its proof are in Appendix A.6.

GPT: R16 — Calibrate the theorem’s contribution and computational meaning. This is an exact Bregman proximal-point guarantee, not a convergence theorem for finite regression, one opponent refresh, a clipped dual, and empirical acceptance. State that distinction beside the headline claim. INPO already studies last-iterate convergence, and EGPO establishes linear convergence for a different regularized equilibrium problem (Zhang et al., 2025; Zhou et al., 2025); their rates are not comparable by exponent alone because targets and access assumptions differ. Here T counts exact inner solves, not labels or neural steps, and all reported LLM configurations use $T = 1$. Show finite-game convergence against total oracle/solver work, including an approximate-solve variant. A useful theoretical extension would relate an inner variational or primal–dual residual to outer error; preserving a last-iterate statement also needs a separate argument, for example control of approximate monotonicity.

Response: R16 — Partially addressed. The headline convergence statement now specifies exact population Bregman proximal solves, defines T as their count, and separates this guarantee from finite fitting and empirical acceptance; the source also states that LLM runs use $T = 1$. A work-normalized outer convergence study and an approximate-solve residual-to-outer-error extension are not supplied.

6 EXPERIMENTS

We examine generated-policy quality, the compromise selected by bargaining, and the effect of adaptive opponents. UltraFeedback (UF-4) supplies a four-objective alignment comparison under score-induced supervision; SafeRLHF supplies a complementary human-pairwise panel. Controlled games isolate the effect of nontransitivity. These panels have different evidential roles: the released UF labels are transitive by construction, and the separate 60,000-judgment audit detects no significant repeated-cycle excess (Appendix C.1). Neither natural panel currently establishes an advantage from cyclic feedback.

6.1 GENERATED-POLICY QUALITY UNDER SCORE-INDUCED SUPERVISION

Protocol and dataset role. The panel uses Llama-3.1-8B-Instruct and UltraFeedback instruction-following (IF), truthfulness, honesty, and helpfulness labels (Cui et al., 2024). Filtering leaves 38,240 eligible prompt groups, supporting disjoint 20k/2k preference-model train/dev, 10k/1k policy train/dev, and 2k final splits. Policies train on newly sampled responses for 1250 updates. UF tests multi-objective alignment and aggregation under a common training protocol; it does not test learning from cyclic labels. Transitivity within each objective does not remove conflict across objectives. Appendix C.3 gives the data, teacher, and training details.

Matched comparisons. Fixed-reference Nash changes opponent adaptation; BT-RM–Nash changes preference representation; game-utilitarian and global game-maxmin change aggregation over the same prompt-averaged game values. The planned PROSPER and MOPO adaptations require faithful implementations of their respective baseline contracts; the main DPO row uses the fixed uniform weight. In particular, PROSPER’s prompt-dependent aggregation does not replace the matched global-maxmin control (Appendix C.3). All comparisons retain their declared feedback and training contracts; L1 coefficient matching does not match realized policy KL.

Evaluation and current results. Every measured policy uses the same planned 2000 final prompts and cached base responses. Table 1 is now computed on the exact intersection: a prompt counts only when every arm in the table has both presentation orders parsed on all four attributes, which leaves 1,644, and each arm’s own count under that same rule is 1,914–1,935. The counts of 1967 and 1960 reported earlier used a different completeness rule and are not the same quantity; they are superseded here rather than carried alongside. Numerical whole-prompt paired intervals are reported for every cell, recomputing the minimum objective inside each replicate, and policy-seed spread is kept separate from prompt uncertainty (Appendix F). No interval on the minimum objective excludes 0.5 for any arm, and the four-objective results do not establish a Nash advantage.

Table 1: UF-4 win rates against the common base, with order averaging and ties worth 0.5. Every entry is computed on the 1,616 prompts of the planned 2,000 for which *all* arms reported here have both presentation orders parsed on all four attributes, so the arms are compared on identical prompts; each method’s own valid count is larger (1,914–1,935) and is reported in Appendix F rather than used here. The common set shrinks as arms are added and is recomputed whenever a row is filled, so these numbers are tied to the set of rows present. $\min_k W_k$ is the minimum attribute mean. Helpfulness is the one attribute whose whole-prompt interval excludes 0.5 for every evaluated seed of every method. Global game-maxmin and fixed-reference Nash, at one seed each, are the only arms whose instruction-following and truthfulness intervals also exclude 0.5 ([0.5084, 0.5412] and [0.5073, 0.5343] for maxmin, [0.5017, 0.5320] and [0.5034, 0.5317] for fixed-reference), and the only arms whose minimum-attribute point estimate exceeds 0.5. Neither minimum’s interval excludes 0.5 ([0.4882, 0.5206] and [0.4892, 0.5212]), and both rest on one policy seed against three for Nash and for utilitarian. Honesty excludes 0.5 for no arm, and the utilitarian honesty seed spread (0.0105) stays several times the others. Interval verdicts move with the row set as much as the point estimates do: on the earlier five-arm common set of 1,719 prompts truthfulness excluded 0.5 for two of three NBPO seeds and now does so for none. Dashes are unmeasured.

Method	Seeds	IF	Truth	Honesty	Help	$\min_k W_k$
Base (defined reference)	—	0.500	0.500	0.500	0.500	0.500
<i>Matched mechanism controls</i>						
NBPO	3	0.5083	0.5108	0.4974	0.5512	0.4974
sample SD over seeds		0.0056	0.0032	0.0031	0.0031	0.0031
Fixed-reference Nash	1	0.5175	0.5173	0.5063	0.5503	0.5063
BT-RM–Nash	—	—	—	—	—	—
Game-utilitarian	3	0.5085	0.5084	0.4975	0.5517	0.4975
sample SD over seeds		0.0021	0.0021	0.0105	0.0023	0.0105
Global game-maxmin	1	0.5249	0.5209	0.5050	0.5600	0.5050
<i>External baselines</i>						
Scalarized DPO (uniform)	—	—	—	—	—	—
PROSPER (adapt.)	—	—	—	—	—	—
MOPO (adapt.)	—	—	—	—	—	—

GPT: R18 — Verify that the HT-MNPO baseline corresponds to the published method. Heterogeneous MNPO trains an interacting population with players associated with different preference oracles (Wu et al., 2026); independent single-objective runs are not automatically the same method. State the player count, oracle assignments, opponent mixture, joint update schedule, and how each reported checkpoint is obtained. Explain what “single-objective HT-MNPO” means in this implementation. Select the “best corner” using a prespecified validation criterion that does not access test results, and report whether any deployment-time mixture is allowed. The relevant distinction is one compromise policy versus a heterogeneous interacting population, not simply multi-objective versus single-objective learning. Consult the August 2026 revision, especially its heterogeneous extension.

Response: R18 — Partially addressed. The revised manuscript no longer presents an unsupported HT-MNPO “best corner” comparison. A faithful heterogeneous-population implementation and its player, opponent, update, and validation-selection contract have not been supplied; no HT-MNPO reproduction claim is made.

6.2 BARGAINING AND EMPIRICAL TRADE-OFFS

On the 1,616 prompts common to all eight evaluated arms, NBPO and the utilitarian control, each over three seeds, obtain helpfulness win rates of 0.5512 and 0.5517, and minimum attribute means of 0.4974 and 0.4975 whose intervals contain 0.5; neither improves all four attributes. Both single-seed controls are higher, and the stronger of the two is global game-maxmin, which reaches 0.5249, 0.5209, 0.5050 and 0.5600 on the four attributes (Table 1). We report this as a negative result for the aggregation rule this paper advocates. Global game-maxmin differs from NBPO in the aggregation rule alone – the same adaptive-game values, the same opponent temperature, a multiplier norm matched to the Nash dual – and it exceeds NBPO’s *best* of three seeds on every attribute and on the minimum, by 2.5 to 3.1 NBPO seed standard deviations. Fixed-reference Nash also exceeds NBPO’s



Figure 1: Empirical trade-off template: helpfulness versus truthfulness (left), and instruction following versus honesty (right). The declared scalarization sweep and uncertainty estimates remain pending.

best seed on instruction following, truthfulness, honesty and the minimum, but not on helpfulness (-0.3 seed standard deviations). Two explanations that would have made this an artifact do not hold. It is not effective step size: the realized Eq. (26) target magnitudes are within 4% across these arms and global game-maxmin’s is the *smallest* of the four (dev RMS 5.42 against NBPO’s 5.47 and fixed-reference’s 5.67). And it is not a representation effect, since global game-maxmin uses the same adaptive-game representation as NBPO. What remains is policy-seed variation: both leading arms are one seed, and seeds 43 and 44 of fixed-reference are running to test it. Restricting to the common prompts matters, and so does which arms are in it: the set has shrunk from 1,719 through 1,644 to 1,616 as arms were added, moving the same NBPO and utilitarian runs from 0.5535 and 0.5527 to 0.5512 and 0.5517 on helpfulness. Their mean target TV is 0.0195; the corresponding SafeRLHF value is 0.0008 (Appendix E). These target diagnostics do not establish policy equivalence. Both matched controls are now measured; the BT-RM, scalarized-DPO, PROSPER and MOPO rows remain unmeasured.

Figure 1 provides two projections for comparing policies with the declared DPO sweep: four specialist vectors, the uniform vector, and two fixed interior vectors. The sweep remains unmeasured. Operating-point selection uses development data, and all seven settings are retained. Dominance is assessed in the full four-dimensional outcome space; the two projections cannot establish it. Worst-objective wins, Nash welfare, and average wins express different preferences over that space, so no single summary establishes the best compromise. Appendix F.4 specifies uncertainty and tuning-cost reporting.

GPT: R21 — Evaluate the compromise rule relative to an attainable trade-off. Average wins, worst-objective wins, and Nash welfare each express a different preference over outcomes; no single one proves that Nash is universally the best compromise. For the two-objective SafeRLHF panel, estimate an empirical trade-off curve using a reasonably tuned weight/constraint sweep and locate each method relative to that curve and the fallback. Use a common validation-only operating-point rule and disclose tuning effort. Combine this with objective-wise results so an aggregation advantage is distinguishable from simply training a stronger policy. The planned training-seed variation measures a different uncertainty source from finite evaluation prompts: report both, using paired prompt resampling and recomputing the worst-objective statistic within each replicate.

Response: R21 — Partially addressed. The main text retains the two-projection trade-off figure and all seven declared DPO weights, while distinguishing full-vector dominance from the chosen compromise statistic. Numerical UF paired intervals, completed sweeps and seed evaluations, and the requested independent SafeRLHF frontier remain pending. The UF plot is not presented as satisfying the SafeRLHF-specific request.

6.3 ADAPTIVE OPPONENTS AND PREFERENCE REPRESENTATION

Controlled nontransitivity. We perturb a transitive preference game while preserving a positive joint-improvement margin. All methods are evaluated under the same adaptive-game values. Figure 2



Figure 2: Controlled nontransitivity: normalized worst surplus versus BT deviance. Points and bars show means and one sample standard deviation over five seeds; all six plotted methods converge.

Table 2: Independent cross-play template. All rows share one evaluator, prompt subset, and comparator bank. Report policy seeds, the common valid prompt count, and paired uncertainty. All outcomes are unmeasured.

Policy	Seeds	N_{common}	$W_{\text{ref}}^{\min}$	$W_{\mathcal{B}}^{\min}$	$\min_k \hat{s}_k^{\mathcal{B}}$
Base	—	—	—	—	—
NBPO	—	—	—	—	—
Fixed-reference Nash	—	—	—	—	—
Dev-selected competitor	—	—	—	—	—

shows that adaptive-game methods remain stable as BT deviance increases, while fixed-reference and scalar-reward methods deteriorate. At the strongest perturbation, normalized worst surplus is 0.931 for NBPO, -0.241 for fixed-reference Nash, and -0.242 for BT-RM-Nash. Game-utitarian (0.908) and Game-KS (0.932) are also strong: the result supports adaptive-game representation, not a unique Nash advantage. The trend replicates on a second construction, with a smaller effect size (Appendix B). The UF-4 panel separates the two claims this figure bundles together, and it does not treat them alike. The representation claim survives: the strongest UF-4 arm is global game-maxmin, which uses the same adaptive-game representation as NBPO. The Nash-aggregation claim does not: on UF-4 maxmin aggregation exceeds NBPO’s best of three seeds on every attribute, at a matched multiplier norm and with a slightly smaller realized target (Table 1). So the controlled family and UF-4 agree that the adaptive-game representation is the load-bearing ingredient, and UF-4 adds that Nash is not the best rule over those values there. The residual tension is narrower than it first appears but real: fixed-reference Nash collapses to -0.241 normalized worst surplus under the strongest controlled perturbation, yet on UF-4 it exceeds NBPO’s best seed on three of four attributes. That is consistent with the released UF labels, transitive within each objective, not departing far enough from a fixed comparator for the controlled failure mode to appear. We do not treat the controlled result as evidence about UF-4 in either direction. Both UF-4 leaders rest on one policy seed against three, with minimum-attribute intervals still containing 0.5; seeds 43 and 44 of the fixed-reference control and the independent cross-play of Table 2 are what would settle it.

Independent natural-policy cross-play. The planned LLM test compares the base, NBPO, fixed-reference Nash, and one development-selected strong competitor on fresh held-out responses. For objective k , $W_k(\pi, q)$ is the independently judged, order-averaged win rate of π against comparator q . With one shared, frozen comparator bank $\mathcal{B}$, we report

$$W_{\text{ref}}^{\min}(\pi) = \min_k W_k(\pi, \pi_{\text{ref}}), \quad W_{\mathcal{B}}^{\min}(\pi) = \min_{k, q \in \mathcal{B}} W_k(\pi, q).$$

The second statistic tests vulnerability to the evaluated bank, not population exploitability. Table 2 separates this comparison from teacher-defined surplus; Appendix F.2 retains the objective-wise matrices and a common regularized bank diagnostic. The natural-policy comparison remains unmeasured.

GPT: R19 — Independent judges and diverse opponents answer different questions. Judge independence is appropriate, but a fixed-reference win rate can miss precisely the exploitable weaknesses used to motivate adaptive opponents. Retain this deployment-oriented metric and add an independently judged cross-play matrix containing the reference, NBPO, fixed-reference Nash, and strong competing policies. Select adversarial comparators on a development pool and evaluate them on disjoint prompts/responses; include candidates outside the four-response training pool. Report a common regularized-game diagnostic under the same independently fixed evaluator as a complementary measure. Otherwise, the experiment can favor fixed-reference training by its metric while never testing whether adaptive training reduces vulnerability to other responses. Robustness and direct reference improvement should be reported as separate outcomes, as R06 demonstrates.

Response: R19 — Partially addressed. Section 6.3 now makes independent cross-play a main mechanism test. Table 2 separates reference wins, worst-bank wins, and a regularized bank diagnostic; Appendix F.2 defines its distinct estimand and objective-wise matrix template. All natural cross-play outcomes remain unmeasured; population exploitability is not claimed.

GPT: R22 — Make the adaptive-opponent falsification test central. The present point estimates favor fixed-reference Nash in average reference wins (.561 versus .556); the appendix also reports higher minimum/average game surpluses (.040/.061 versus .028/.056). First confirm that these game surpluses use the same evaluator, temperature, and prompt set for both policies. The draft already acknowledges this negative result and proposes diagnostics; make those diagnostics the main mechanism experiment. Solve both targets accurately on identical finite matrices at the actual finite temperature, then measure the loss incurred by neural realization. Add a controlled transitive-to-cyclic family while maintaining joint feasibility, and evaluate diverse held-out comparators. The aim is to identify a regime where adaptivity helps, or establish why it does not, not to require NBPO to win every setting.

Response: R22 — Partially addressed. The controlled figure is moved to the main mechanism subsection, with Game-utilitarian and Game-KS retained to separate representation from Nash-specific claims. A matched natural-policy cross-play summary and full objective-wise template are added. The LLM comparison remains unmeasured, and historical contrary results are not treated as reconciled.

6.4 CAPABILITY, REALIZATION, AND SUPPORTING RESULTS

Capability retention. Table 3 reserves core capability and safety outcomes for the same UF-trained checkpoints. IFEval and GSM8K are now measured on all three seeds of NBPO and of game-utilitarian. GSM8K is unchanged: every seed is within 0.008 of the base, and no per-seed paired prompt-bootstrap interval on the base-relative difference excludes zero. IFEval declines for both methods, by 0.0277 for NBPO and 0.0290 for game-utilitarian at the seed mean, with the per-seed paired intervals excluding zero for two of three NBPO seeds and for none of the game-utilitarian seeds. The decline is therefore a property of this 250-update training regime rather than of the aggregation rule, and it does not separate the two rules. The safety columns remain pending. The completed ARC-C/HellaSwag changes are at most 0.0030 from the base, and the MMLU changes are -0.0007 for NBPO and -0.0006 for game-utilitarian; these point estimates do not establish equivalence. The full capability table, including the separate historical SafeRLHF protocol, remains in Appendix F.1.

GPT: R20 — Explain the existing capability regression and benchmark score definitions. Table [original-draft ref.: `tab:general-benchmark-completion`] reports a Llama IFEval decline from .762 to .701, or 6.1 percentage points, despite instruction following appearing in the training panel. This is a concrete transfer failure worth analyzing, not something that an average judged win rate resolves. Inspect failures by constraint type, answer length, and refusal behavior, and test a matched instruction-following/length control. Also distinguish official benchmark protocols from custom pairwise judging on benchmark prompts: converting MT-Bench or other benchmarks into reference-policy wins changes the metric and should be labeled explicitly. For small differences in judged scores, use paired prompt-level uncertainty and report evaluator disagreement rather than relying on three-decimal point estimates.

Response: R20 — Partially addressed. The main capability template now targets the same UF checkpoints, with the complete measured ARC-C/HellaSwag/MMLU and historical SafeRLHF panel

Table 3: Core capability template for the UF-4 policies. IFEval uses strict-prompt accuracy; GSM8K uses exact match. HB is HarmBench harmful-completion rate, and XS is XSTest benign refusal. NBPO and game-utilitarian entries are means over the same three policy seeds, with the sample standard deviation beneath. The official IFEval language check calls a seed-unpinned language detector, so rescoring one fixed response set moves strict-prompt accuracy by up to 2/541; the base row is its frozen first scoring. GSM8K exact match is deterministic. Dashes are unmeasured.

Method	Seeds	IFEval	GSM8K	HB↓	XS↓
Base	1	0.7542	0.8666	0.2813	—
NBPO	3	0.7264	0.8633	—	—
<i>sample SD over seeds</i>		0.0098	0.0031		
Fixed-reference Nash	—	—	—	—	—
BT-RM–Nash	—	—	—	—	—
Game-utilitarian	3	0.7252	0.8625	—	—
<i>sample SD over seeds</i>		0.0059	0.0034		
Global game-maxmin	—	—	—	—	—
Scalarized DPO (uniform)	—	—	—	—	—
PROSPER (adapt.)	—	—	—	—	—
MOPO (adapt.)	—	—	—	—	—

preserved in Appendix F.1. Trained-policy core capability/safety outcomes, matched length controls, and the historical IFEval-drop reconciliation remain pending.

Solver, realization, and cost. Recomputation from stored UF tensors gives a dual residual of 3.74×10^{-14} and maximum inner stationarity spread 1.88×10^{-10} . Across three policy seeds, held-out neural nMSE is 0.5861 ± 0.0033 , compared with 1 for the unchanged policy. This measures fitting to the frozen target, not generated-policy quality. The reported seed-42 projection takes 11.2 GPU-hours; it excludes other pipeline stages. Appendix E gives diagnostics, and Appendix F.3 separates feedback, generation, teacher, solver, training, evaluation, and tuning costs.

GPT: R17 — Match information and computation, not only neural step counts. The closest-method comparison needs separate accounting for unique semantic comparisons, order-swapped judge calls, generated response tokens, neural training compute, and hyperparameter trials. Equalizing 300 neural updates does not equalize the cost of candidate generation, opponent estimation, or repeated policy fitting. Report a fixed-label-budget comparison and a performance-versus-total-compute curve for NBPO, fixed-reference Nash, and the strongest preference-based competitors. For reward-model methods, use the same underlying preference information where feasible, charge reward fitting to their budget, and disclose extra pretrained supervision. If cached comparisons are reused across many numerical iterations, explain that reuse rather than counting every iteration as new evidence. These controls determine whether the optimizer is practically competitive.

Response: R17 — Partially addressed. The main text reports the measured projection cost, and Appendix F.3 adds a method-wise accounting template for feedback, teacher work, policy training, generation/evaluation, and tuning. The 11.2 GPU-hours cover only UF seed-42 projection. Matched total-budget results and performance-versus-compute curves remain unmeasured.

Supporting helpfulness–safety comparison. SafeRLHF supplies human helpfulness and harmlessness comparisons on the same response pairs. Its completed 250-update comparison uses 2000/500/1000 policy train/dev/hold-out prompts and three policy seeds. Nash and L1-matched utilitarian have positive teacher-defined aggregate surplus, without a resolved Nash advantage (Table 20). These values are not independent policy outcomes. SafeRLHF IFEval means are 0.748 and 0.745, against a historical base value of 0.752 under a protocol still requiring reconciliation. Teacher truncation affects 26,790/84,000 candidate occurrences. The requested independent helpfulness–safety frontier remains pending; UF’s four-objective plot does not replace it.

GPT: R23 — The current sign test is weaker than invariance of the implemented policy. Specify whether the experiment rescales completed values or primitive payoffs; R08 gives the required temperature transformation for the latter. Exact dual weights transform as $\lambda'_k = \lambda_k/c_k$. Matching finite projected-gradient trajectories additionally requires transformed initial weights, coordinate bounds, floors, and coordinate step sizes $\gamma'_k = \gamma_k/c_k^2$. Compare full target magnitudes, opponent distributions, finite-pool policy KL, and neural outcomes, not only signs. Explain the identical utilitarian, Uniform-INPO, and MOPO rows: if they use a common frozen-score surrogate rather than each method’s actual update, label that restriction. Do not infer algorithm-level scale sensitivity from an undocumented surrogate. Near-zero scores should be separated from substantive direction changes, and numerical covariance should not be described as robustness to arbitrary judge miscalibration.

Response: R23 — Partially addressed. The appendix now states the required payoff–temperature transformation and the corresponding scaling of multipliers, initialization, bounds, thresholds, and coordinate step sizes; it also states that target signs alone are insufficient. The earlier sign-only comparison is no longer used as evidence, but full-target, opponent, finite-pool-policy, and neural covariance measurements remain pending.

7 RELATED WORK

General pairwise preferences. NLHF and its scalable variants optimize directly against general pairwise preference models rather than fitting scalar rewards (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Blackwell-style methods extend this view to vector-valued feedback and target sets (Blackwell, 1956; Bhatia et al., 2020). PROSPER introduces a maximum-entropy Blackwell winner with a scalable single-policy update and an inner worst-case choice over objective weights and opponents (Zhang et al., 2026). NBPO instead keeps one regularized game value per objective and applies Nash bargaining across the improvements over a shared fallback.

Multi-objective alignment.

GPT: R24 — Separate the aggregation rule from the order of aggregation over prompts. PROSPER’s adversarial objective weights are prompt-dependent, whereas NBPO’s inverse-surplus weights operate on objective values averaged over prompts (Zhang et al., 2026). Thus the comparison changes both the compromise criterion and the ability to compensate for poor performance on one prompt with improvement on another. Write the two objectives side by side and include a two-prompt example showing the distinction. Retain a faithful prompt-wise PROSPER baseline and the matched global-maxmin control over NBPO’s game values. Without this separation, an empirical difference cannot be attributed specifically to Nash bargaining. Make clear that the individual-rationality guarantee is about population-average objective game values, not every prompt or every stakeholder.

Response: R24 — Partially addressed. The main results retain global game-maxmin alongside faithful PROSPER: the former changes the compromise over shared prompt-averaged values, whereas the latter also changes aggregation across prompts. The appendix keeps the two-prompt example and formal distinction. Their matched policy outcomes remain unmeasured.

MODPO, Rewarded Soups, and Panacea expose trade-offs through scalarization, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024). MaxMin-RLHF and RMOD optimize weakest-objective criteria (Chakraborty et al., 2024; Son et al., 2026). MOPO uses pairwise preferences without point-wise rewards, selecting a primary objective and lower-bound constraints with a lagged reference in its practical algorithm (Agnihotri et al., 2026). RACO resolves conflicts between DPO-style objective gradients (Chen et al., 2026). These approaches differ from bargaining over reference-relative robust policy values.

Social choice and bargaining.

GPT: R25 — Correct the substantive classification of MOPO and discuss RACO. The preceding paragraph incorrectly includes MOPO among methods requiring scalar objective rewards or transitive preference models. MOPO explicitly uses pairwise preferences without a point-wise reward assumption; its June 2026 revision also refreshes a lagged reference and constraint schedule (Agnihotri et al., 2026). Distinguish its primary-objective/constraint selection and reference-based comparisons from NBPO’s symmetric surplus bargaining and policy-dependent adversarial opponent. Also position RACO, a recent reward-free method for conflicting objectives using preference-loss updates (Chen et al., 2026); reward-free training alone is not new. A matched RACO comparison is useful, but distinguish its DPO-style representation from unrestricted cyclic preference modeling. This correction is necessary for a credible novelty argument, not merely broader citation coverage.

Response: R25 — Addressed. Related Work now describes MOPO as using pairwise preferences without point-wise rewards, with a primary objective, lower-bound constraints, and a lagged reference. It also positions RACO through DPO-style objective gradients, so reward-free training alone is not presented as NBPO’s novelty; a matched RACO experiment is not reported.

The Nash solution and its Pareto, uniqueness, and affine-invariance properties are classical (Nash, 1950). Social-choice work argues that alignment should state its aggregation rule explicitly (Conitzer et al., 2024). Multi-party RLHF studies utilitarian, Nash, and leximin welfare over party-specific learned rewards and separately considers reward-free von Neumann solutions (Zhong et al., 2026); Nash bargaining also selects immediate task-gradient improvements in multi-task learning (Navon et al., 2022). Our contribution is not a new bargaining axiom, but a formulation and practical optimizer for applying the classical Nash rule to policy-dependent objective-wise preference-game values.

8 CONCLUSION AND LIMITATIONS

GPT: R26 — Compare against the current Nash-welfare literature at the level of the optimized object. The September 7, 2026 revision of Zhong et al., now titled *Provable Pluralistic Alignment: Multi-Party RLHF under Offline Human Feedback*, includes disagreement-adjusted bargaining and separately treats general preferences through a von Neumann formulation. Update the bibliography/positioning rather than relying on the older title. Explain that its welfare analysis concerns learned party rewards, Nash-MTL bargains over immediate gradient improvements, and NBPO bargains over robust policy-value surpluses. Nash welfare, inverse-improvement weighting, and reference subtraction are therefore not individually sufficient novelty. Besides fixed-reference Nash, fit scalar rewards to the same comparisons and apply Nash aggregation as a controlled baseline; this isolates the incremental value of the game representation without confounding it with the welfare choice.

Response: R26 — Partially addressed. The positioning distinguishes learned party rewards, reward-free von Neumann solutions, immediate gradient improvements, and robust policy-value surpluses. Table 5 now includes controlled BT-RM–Nash results; its matched natural-language comparison remains unmeasured. The cited bibliography revision still requires verification.

NBPO applies Nash bargaining to reference-relative preference-game improvements through a constrained proximal solver. Controlled experiments support adaptive-game representations. UltraFeedback tests standard multi-objective alignment under score-induced supervision and shows measurable neural realization; it does not supply cyclic training feedback. Neither its single-seed independent policy comparison nor the three-seed teacher-defined SafeRLHF study establishes a Nash advantage over L1-matched utilitarian aggregation. The separate natural-preference audit detects no repeated-cycle excess. Matched fixed-reference and external-baseline comparisons, independent cross-play, and empirical trade-off assessment remain incomplete. Joint improvement may be infeasible, and exact population guarantees do not certify a one-stage neural policy or per-prompt safety.

AI USE STATEMENT

In this work, generative AI tools were used to critique the problem formulation, assist with mathematical derivations and proof presentation, propose experimental checks, and edit manuscript text. The authors are responsible for independently verifying every theorem, citation, implementation, empirical result, and disclosure in the final submission.

ETHICS STATEMENT

This work studies aggregation rules for language-model alignment, including harmlessness as one objective. Improvement in a game value is not an absolute safety certificate. Safety-critical deployment should retain hard constraints and independent safety evaluations in addition to the bargaining objective.

REPRODUCIBILITY STATEMENT

Algorithm 1 distinguishes the population proximal problem, the numerically solved finite-pool inner problem, and its neural projection. Appendix A contains the dual and convergence proofs; Appendices B–F specify constructions, supervision, hyperparameters, diagnostics, and evaluation protocols. The submission package should include candidate pools, preference tensors, dual traces, solver residuals, per-seed outputs, and the code and configurations required to reproduce the reported results.

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A PROOFS

A.1 PROOF OF PROPOSITION 1

For a fixed comparator ν , define

$$\mathcal{J}_{k,\nu}(\pi) := J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)).$$

The first term is linear in π and the second does not depend on π , so $\mathcal{J}_{k,\nu}$ is affine. The pointwise minimum of affine functions is concave; hence V_{k,β_k} is concave. Since $|\Delta_k| \leq 1/2$,

$$|V_{k,\beta_k}(\pi) - V_{k,\beta_k}(\pi')| \leq \frac{1}{2} \mathbb{E}_{x \sim \mathcal{D}} \|\pi(\cdot | x) - \pi'(\cdot | x)\|_1,$$

which also proves continuity.

For $\beta_k > 0$, the inner minimization separates across prompts. Fix x and abbreviate $m(z) = m_k^\pi(z | x)$. The prompt-wise problem is

$$\min_{\nu \in \Delta(\mathcal{Y}_x)} \sum_z \nu(z) m(z) + \beta_k \sum_z \nu(z) \log \frac{\nu(z)}{\pi_{\text{ref}}(z | x)}.$$

The first-order condition under $\sum_z \nu(z) = 1$ gives

$$\nu(z) \propto \pi_{\text{ref}}(z | x) \exp\{-m(z)/\beta_k\},$$

which yields (8); substituting it back yields (9). The KL term makes the minimizer unique. Danskin’s theorem therefore gives, for every feasible direction h satisfying $\sum_y h(y | x) = 0$,

$$DV_{k,\beta_k}(\pi)[h] = \mathbb{E}_{x \sim \mathcal{D}} \sum_y h(y | x) \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} \Delta_k(y, z | x).$$

Thus the coordinate derivative is

$$[\nabla_\pi V_{k,\beta_k}(\pi)]_{x,y} = \mathcal{D}(x) Q_k^\pi(y | x),$$

proving (10).

A.2 CONNECTION TO THE FIXED-REFERENCE

This subsection gives a finite-action illustration at the unregularized endpoint $\beta = 0$. It is not used in Assumption 1, the full-support analysis, or the finite-temperature experiments.

Define the fixed-reference improvement

$$a_k(\pi) = J_k(\pi, \pi_{\text{ref}}) = P_k(\pi \succ \pi_{\text{ref}}) - \frac{1}{2}. \quad (28)$$

Skew-symmetry gives $a_k(\pi_{\text{ref}}) = 0$.

Full cyclic example. For one prompt, let $\pi_{\text{ref}} = \delta_r$, where δ_y is the deterministic policy that always outputs y . Suppose A , B , and C each beat r with probability 0.75 and form the deterministic cycle

$$A \succ B, \quad B \succ C, \quad C \succ A.$$

Both

$$\pi_{\text{pure}} = \delta_A \quad \text{and} \quad \pi_{\text{mix}} = \frac{\delta_A + \delta_B + \delta_C}{3}$$

have the same fixed-reference improvement,

$$a(\pi_{\text{pure}}) = a(\pi_{\text{mix}}) = 0.75 - \frac{1}{2} = 0.25.$$

Their worst-opponent values are nevertheless different. For the pure policy, response C defeats A deterministically, so

$$V_0(\pi_{\text{pure}}) = \min_{\nu} J(\pi_{\text{pure}}, \nu) = J(\delta_A, \delta_C) = -\frac{1}{2}.$$

For the uniform mixture, consider first the pure opponent δ_A . The mixture outputs A , B , and C with equal probability. Against A , these responses respectively tie, lose, and win, giving centered payoffs 0, $-1/2$, and $+1/2$. Hence

$$J(\pi_{\text{mix}}, \delta_A) = \frac{1}{3} \left(0 - \frac{1}{2} + \frac{1}{2} \right) = 0.$$

By symmetry,

$$J(\pi_{\text{mix}}, \delta_B) = J(\pi_{\text{mix}}, \delta_C) = 0.$$

Against the reference,

$$J(\pi_{\text{mix}}, \delta_r) = 0.25,$$

because every response in the mixture beats r with probability 0.75. Since $J(\pi_{\text{mix}}, \nu)$ is linear in ν , no mixed opponent can achieve a value below the minimum over these pure opponents. Therefore

$$V_0(\pi_{\text{mix}}) = \min_{\nu} J(\pi_{\text{mix}}, \nu) = \min\{0, 0, 0, 0.25\} = 0.$$

Finally, the reference itself has

$$V_0(\pi_{\text{ref}}) = \min_{\nu} J(\delta_r, \nu) = -0.25,$$

since r loses to each of A , B , and C with probability 0.75. Thus

	$a(\pi)$	$V_0(\pi)$	$V_0(\pi) - V_0(\pi_{\text{ref}})$
π_{pure}	0.25	-0.50	-0.25
π_{mix}	0.25	0	0.25

The fixed reference therefore assigns the two policies the same score, even though the pure policy is fully exploitable by one response whereas the balanced mixture guarantees an average tie against every response in the cycle.

Game-value improvement need not imply a direct reference win. For one prompt, consider

$$A = \begin{pmatrix} 0 & .5 & -.5 \\ -.5 & 0 & .5 \\ .5 & -.5 & 0 \end{pmatrix}, \quad \mu = (.8, .1, .1), \quad p = (.25, .43, .32), \quad \beta = .25.$$

Both policies have full support. Direct evaluation gives $V_{\beta}(\mu) = -.0517901$ and $V_{\beta}(p) = -.0417293$, hence $s(p) = .0100608 > 0$. Yet $p^{\top} A \mu = -.0385$, so the direct reference win probability is .4615. Robustness-adjusted value and direct wins answer different questions even under the same judge.

A.3 POLICY-INDUCED NASH BARGAINING AND CLASSICAL CONSEQUENCES

We prove Proposition 2. Since Π is compact and each V_{k,β_k} is continuous, $V_\beta(\pi)$ is bounded. The definition of $\mathcal{G}_\beta$ then implies boundedness. Closedness follows by taking any convergent sequence $v_n \in \mathcal{G}_\beta$, choosing policies π_n with $d \leq v_n \leq V_\beta(\pi_n)$, extracting a convergent policy subsequence, and applying continuity. Downward closure within $\{v : v \geq d\}$ is immediate from the definition.

For convexity, take $v^1, v^2 \in \mathcal{G}_\beta$ with witnesses π_1, π_2 . Objective-wise concavity from Proposition 1 gives, for $\alpha \in [0, 1]$,

$$V_\beta(\alpha\pi_1 + (1 - \alpha)\pi_2) \geq \alpha V_\beta(\pi_1) + (1 - \alpha)V_\beta(\pi_2)$$

componentwise, so $\alpha v^1 + (1 - \alpha)v^2 \in \mathcal{G}_\beta$. Assumption 1 provides a point strictly above d , proving nondegeneracy.

For any $v \in \mathcal{G}_\beta$ with $v > d$, some policy π satisfies $v \leq V_\beta(\pi)$. Since the Nash objective is coordinatewise increasing,

$$\sum_k \log(v_k - d_k) \leq \sum_k \log(V_{k,\beta_k}(\pi) - d_k) = F(\pi).$$

Conversely, every $\pi \in \Pi_+$ induces a feasible value vector $V_\beta(\pi)$. Hence (12) realizes the classical Nash solution over $(\mathcal{G}_\beta, d)$.

The remaining statements are standard Nash-bargaining consequences, but we record the short arguments for completeness. Since the log objective tends to $-\infty$ when any surplus approaches zero, every optimizer lies strictly above d . Coordinatewise monotonicity implies Pareto optimality. Strict concavity of $\sum_k \log s_k$ over value vectors implies that all optimal policies induce the same surplus, hence the same game-value vector, although the policy itself can be nonunique.

Finally, under independent positive affine transformations

$$V'_k(\pi) = c_k V_k(\pi) + b_k, \quad d'_k = c_k d_k + b_k, \quad c_k > 0,$$

the transformed surplus is $s'_k(\pi) = c_k s_k(\pi)$ and

$$\sum_k \log s'_k(\pi) = \sum_k \log s_k(\pi) + \sum_k \log c_k.$$

The added term is independent of π , so the optimizer set is unchanged.

Table 4: Guarantees of the stated population rules under joint improvement. A “no” means the property need not hold for every optimizer without an additional tie-break. This is not an exhaustive comparison of bargaining solutions.

rule	strict IR	Pareto optimal	unique value	affine invariant
positive-weight utilitarian	no	yes	no	no
absolute maxmin	no	no	no	no
surplus maxmin	yes	no	no	no
Nash bargaining	yes	yes	yes	yes

Scope of invariance and alternatives. Affine invariance transforms completed utilities and disagreement values together. At the primitive level, $A'_k = c_k A_k$ requires $\beta'_k = c_k \beta_k$ to preserve the opponent and obtain $V'_k = c_k V_k$. With fixed temperature instead, $V_{cA,\beta} = cV_{A,\beta/c}$. If scaled margins leave $[-1/2, 1/2]$, the transformation changes payoff units rather than defining a new probability oracle. Arbitrary judge recalibration is not covered.

Exact dual multipliers transform as $\lambda'_k = \lambda_k/c_k$, leaving $\lambda'_k Q'_k = \lambda_k Q_k$ and the pair target unchanged. Matching projected-gradient trajectories further requires scaling initial multipliers and coordinate bounds by $1/c_k$, any diagnostic surplus thresholds by c_k , and coordinate step sizes by $1/c_k^2$. Target signs alone do not verify these stronger identities. We do not floor surpluses to make infeasible targets positive.

A Pareto-refining tie-break can remove the dominated-optimum failure of raw maxmin. Relative-improvement bargaining provides another scale-invariant choice but requires objective-wise ideal values and, in general, an appropriate refinement (Kalai & Smorodinsky, 1975; Han et al., 2026). These alternatives do not make Nash uniquely admissible. Here the bargaining parties are objective rubrics: duplicating a rubric duplicates its log-surplus term. The fixed reference also plays two distinct roles, disagreement policy and opponent anchor; changing both changes the utilities as well as disagreement.

Rubric and reference sensitivity (pending). **TO-DO:** Prespecify a duplication or merge of an overlapping rubric and report its effect on the selected game values and policy outcomes. Separately vary the disagreement policy while holding the opponent anchor fixed, and vary the opponent anchor while holding the disagreement policy fixed. Recompute the disagreement values under each corresponding game definition; a fixed disagreement policy need not have fixed game values when the anchor changes. These checks assess modeling choices that affine invariance does not remove.

A.4 COUNTEREXAMPLES FOR ALTERNATIVE AGGREGATION RULES

GPT: R27 — Compare with a strong bargaining alternative, not only unnormalized maxmin. Table 4 characterizes the particular rules as defined; it does not establish that Nash is the only reasonable scale-invariant compromise. The Pareto counterexample for maxmin exploits a dominated member of an optimum set, so a Pareto-refining tie-break can remove that failure. Consider a normalized relative-improvement maxmin or Kalai–Smorodinsky control, using each objective’s attainable improvement above the fallback; Han et al. (2026) is a directly relevant 2026 treatment, and Kalai & Smorodinsky (1975) supplies the classical alternative. Explain the additional ideal-point estimation cost and use validation data fairly. Comparing Nash with this stronger alternative would test the substantive bargaining choice; demonstrating that unnormalized maxmin depends on units mainly confirms a known algebraic property.

Response: R27 — Partially addressed. The rule comparison acknowledges Pareto-refining maxmin tie-breaking and relative-improvement bargaining, including the cost of ideal-value estimation. Table 5 now reports a Game-KS control, so Nash is not presented as uniquely admissible or empirically superior. A matched natural-language KS or normalized-maxmin comparison remains pending.

The positive guarantees in Table 4 follow directly from the corresponding objectives. Because $w_k > 0$, the positive-weight utilitarian objective $\sum_k w_k V_k(\pi)$ is strictly increasing in every game-value coordinate, so none of its optimizers can be Pareto dominated. Under Assumption 1, surplus maxmin also attains a strictly positive minimum surplus because a jointly improving policy exists. The examples below establish the guarantees that can fail for the alternative rules.

All examples use three responses, so a policy is a probability vector $\pi = (\pi_1, \pi_2, \pi_3) \in \Delta_3$, and use the uniform reference $\pi_{\text{ref}} = (1/3, 1/3, 1/3)$. For objective k , Δ_k is the skew-symmetric pairwise payoff matrix: $(\Delta_k)_{ij}$ is the centered preference margin of response i against response j . At $\beta_k = 0$,

$$V_k(\pi) = \min_j (\pi^\top \Delta_k)_j, \quad d_k = V_k(\pi_{\text{ref}}), \quad s_k(\pi) = V_k(\pi) - d_k.$$

Thus $\pi^\top \Delta_k$ contains the expected payoff of π against each pure opponent, and $V_k(\pi)$ is its smallest coordinate.

Because each V_k is the minimum of linear functions, the optimizers below are obtained from small linear programs. For example, equal-weight utilitarian optimization can be written as

$$\max_{\pi, t_1, t_2} t_1 + t_2 \quad \text{s.t.} \quad t_k \leq (\pi^\top \Delta_k)_j \quad \forall k, j, \quad \pi \in \Delta_3.$$

Absolute maxmin uses one variable t with $t \leq (\pi^\top \Delta_k)_j$, while surplus maxmin uses $t + d_k \leq (\pi^\top \Delta_k)_j$ for every k, j .

Utilitarian failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \\ -1/4 & 1/2 & 0 \end{pmatrix}.$$

1296 For the uniform reference,
1297 $\pi_{\text{ref}}^\top \Delta_1 = (1/3, 0, -1/3), \quad \pi_{\text{ref}}^\top \Delta_2 = (1/12, 0, -1/12),$
1298
1299 and hence
1300 $d = (-1/3, -1/12).$
1301
1302 First, joint improvement is feasible. The policy
1303 $\bar{\pi} = (1/4, 1/4, 1/2)$
1304
1305 is only a feasibility witness; it gives
1306 $\bar{\pi}^\top \Delta_1 = (3/8, 1/8, -1/4), \quad \bar{\pi}^\top \Delta_2 = (0, 1/8, -1/16).$
1307
1308 Therefore
1309 $V(\bar{\pi}) = (-1/4, -1/16), \quad s(\bar{\pi}) = V(\bar{\pi}) - d = (1/12, 1/48) > 0.$
1310
1311 In contrast, solving the equal-weight utilitarian problem $\max_{\pi} V_1(\pi) + V_2(\pi)$ gives
1312 $\pi_U = (0, 1/5, 4/5).$
1313
1314 For this policy,
1315 $\pi_U^\top \Delta_1 = (1/2, 2/5, -1/10), \quad \pi_U^\top \Delta_2 = (-1/10, 2/5, -1/10),$
1316
1317 so
1318 $V(\pi_U) = (-1/10, -1/10), \quad s(\pi_U) = (7/30, -1/60).$
1319
1320 Thus joint improvement is possible, yet a utilitarian optimum falls below the fallback on objective 2.

1321 **Absolute-maxmin failure of strict improvement.** Let
1322
1323 $\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & 1/2 \\ -1/4 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 1/4 & -1/2 \\ -1/4 & 0 & 1/2 \\ 1/2 & -1/2 & 0 \end{pmatrix}.$
1324
1325
1326 Again,
1327 $d = (-1/3, -1/12).$
1328
1329 Joint improvement is feasible, as witnessed by
1330 $\bar{\pi} = (1/3, 7/15, 1/5),$
1331
1332 for which
1333 $V(\bar{\pi}) = (-4/15, -1/60)$
1334
1335 and therefore
1336 $s(\bar{\pi}) = (1/15, 1/15) > 0.$
1337
1338 Absolute maxmin instead solves
1339 $\max_{\pi} \min\{V_1(\pi), V_2(\pi)\},$
1340
1341 whose linear-program solution is
1342 $\pi_{\text{AM}} = (0, 4/5, 1/5).$
1343
1344 For this policy,
1345 $\pi_{\text{AM}}^\top \Delta_1 = (7/20, -1/10, 2/5), \quad \pi_{\text{AM}}^\top \Delta_2 = (-1/10, -1/10, 2/5),$
1346
1347 so
1348 $V(\pi_{\text{AM}}) = (-1/10, -1/10)$
1349
1350 and
1351 $s(\pi_{\text{AM}}) = (7/30, -1/60).$
1352
1353 Thus absolute maxmin maximizes the smallest *absolute* game value while allowing objective 2 to fall below its own fallback.

A Pareto-dominated surplus-maxmin optimum. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}.$$

The uniform reference gives

$$d = (-1/3, -1/6).$$

For objective 2, no policy can achieve $V_2(\pi) > 0$, while every mixture supported on the first two responses achieves $V_2(\pi) = 0$. Consequently,

$$s_2(\pi) \leq 1/6$$

for every policy, so the optimal surplus-maxmin value cannot exceed $1/6$.

The two policies

$$\pi_c = (1/3, 2/3, 0), \quad \pi_e = (0, 1, 0)$$

satisfy

$$V(\pi_c) = (-1/6, 0), \quad s(\pi_c) = (1/6, 1/6),$$

and

$$V(\pi_e) = (0, 0), \quad s(\pi_e) = (1/3, 1/6).$$

Both therefore maximize

$$\min_k s_k(\pi)$$

at $1/6$. However,

$$s(\pi_e) > s(\pi_c)$$

in the first coordinate and is equal in the second, so π_e Pareto dominates π_c . Thus the surplus-maxmin optimizer set can contain Pareto-dominated policies.

A.5 DUAL DERIVATION, DIRECT SOLVE, AND NEURAL PROJECTION

For multipliers $\lambda_k \geq 0$, the Lagrangian of (16) is

$$\mathcal{L}_t(\pi, u, \lambda) = \sum_k \log u_k + \sum_k \lambda_k (s_k(\pi) - u_k) - \eta_t^{-1} D(\pi \| \pi_t).$$

For fixed $\lambda_k > 0$, maximizing $\log u_k - \lambda_k u_k$ gives $u_k = 1/\lambda_k$ and value $-\log \lambda_k - 1$; if $\lambda_k = 0$, the maximum is unbounded. Substitution yields (17), and the remaining maximization over π is (18). Uniqueness of that maximizer and Danskin's theorem yield (19).

Assumption 1 provides strict feasibility: choose $\bar{\pi}$ with $s_k(\bar{\pi}) > 0$ and $0 < \bar{u}_k < s_k(\bar{\pi})$. Slater's condition therefore gives strong duality. Stationarity in u_k gives

$$\lambda_{t+1,k} = \frac{1}{u_{t+1,k}} > 0,$$

and complementary slackness then forces $u_{t+1,k} = s_k(\pi_{t+1})$. This proves Proposition 3.

Dual step used as a fallback. Stationarity $\hat{s}_k(p_t^*(\lambda)) = 1/\lambda_k$ can also be approached by projected subgradient *descent* on the dual — the update below subtracts the dual gradient, and the dual of the concave inner problem is minimized over λ —

$$\lambda^{(m+1)} = \Pi_\Lambda \left[\lambda^{(m)} - \gamma_m \left(\hat{s} \left(p_t^*(\lambda^{(m)}) \right) - (\lambda^{(m)})^{-1} \right) \right], \quad (29)$$

where $\gamma_m > 0$ is the dual step size, Π_Λ is Euclidean projection onto $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$ with $0 < \lambda_{\min} < \lambda_{\max}$, and the inverse is elementwise. On the pools reported here this iteration stalls at a projected KKT residual near 10^{-2} after 3000 steps, whereas the root solve reaches 10^{-9} in tens of inner solves; no reported result uses (29).

What the projected certificate certifies. The projected residual concerns the box-constrained dual. Original stationarity additionally requires $\|\hat{s} - \lambda^{-1}\|_\infty$ to meet tolerance; active bounds cannot be ignored merely because projection removes the residual. We record each bound hit and classify a solve with unresolved unprojected stationarity as uncertified. Interior numerical solutions with small residuals support the finite problem only. The supplied SafeRLHF report gives $\Lambda = [10^{-3}, 10^3]^2$, $\lambda = (6.33, 6.19)$, and independently recomputed unprojected residual 6.7×10^{-11} . These are teacher diagnostics, not neural-policy certificates.

Finite-pool optimality and neural projection. Fix λ and let $\pi_t^*(\lambda)$ solve (18). First-order optimality gives, for every prompt and response pair in the support,

$$h_t(\pi_t^*(\lambda); x, y, y') = \eta_t \sum_k \lambda_k \left(Q_k^{\pi_t^*(\lambda)}(y | x) - Q_k^{\pi_t^*(\lambda)}(y' | x) \right). \quad (30)$$

The direct finite-pool program (23) is the sampled analogue of this strictly concave problem. Its normalized optimizer determines all finite-pool log-probability differences, so (24) is identified without computing a language-model normalizer. Equation (27) is the corresponding finite-pool KKT identity and is checked numerically before an artifact is accepted. If the positive-probability pair graph connects all sampled occurrences, zero regression error implies

$$\tilde{p}_{\theta,x}(i) = \frac{p_{t,x}(i) \pi_\theta(y_{x,i} | x) / \pi_t(y_{x,i} | x)}{\sum_j p_{t,x}(j) \pi_\theta(y_{x,j} | x) / \pi_t(y_{x,j} | x)} = p_{t,x}^*(i).$$

This identifies the importance-reweighted occurrence measure only. For example, under a uniform three-response anchor, fitting only pair (1, 2) with target zero permits every neural distribution $(a, a, 1 - 2a)$, $0 < a < 1/2$. Unseen-response mass remains unconstrained. Resampling a new learner pool at each outer stage makes the empirical base measure track π_t ; reusing a pool from another proposal instead requires explicit importance weights. The reference opponent anchor remains π_{ref} .

Why an optimal fixed point need not be reached. Use the cyclic matrix A above, reference and proximal center $\mu = (.34, .33, .33)$, $\beta = .25$, and fixed $\eta\lambda = 4$. Undamped best-response alternation applies

$$T(p) = \text{softmax}(\log \mu + 4A \text{softmax}(\log \mu + Ap/\beta)).$$

Its fixed point is $p^* = (.33618258, .32907171, .33474571)$, but both tangent-space Jacobian eigenvalues are approximately -1.333163 . Perturbations therefore alternate and expand locally. This is not excluded by improvement: $V(p^*) - V(\mu) \simeq 4.7194 \times 10^{-5} > 0$. In the supplied deterministic check, 100 iterations from μ leave $\|p - p^*\|_2 \simeq .27986$; a direct constrained solve of the same fixed-multiplier problem reaches inner stationarity below 10^{-9} . These are finite-game calculations, not LLM results or a complete Nash dual solve.

For general fixed λ , let $f_x(p; \lambda)$ denote the objective in (23) and let $\nu_k(p) = \text{softmax}(\log \mu_x - \hat{A}_{k,x}^\top p / \beta_k)$. On the simplex interior,

$$\begin{aligned} \nabla f_x(p; \lambda) &= \sum_k \lambda_k \hat{A}_{k,x} \nu_k(p) - \eta_t^{-1} (\log(p/p_{t,x}) + \mathbf{1}), \\ \nabla^2 f_x(p; \lambda) &= - \sum_k \frac{\lambda_k}{\beta_k} \hat{A}_{k,x} (\text{diag}(\nu_k) - \nu_k \nu_k^\top) \hat{A}_{k,x}^\top - \eta_t^{-1} \text{diag}(1/p) \prec 0. \end{aligned}$$

Thus the finite target is unique, and direct optimization includes opponent feedback through this curvature. Strict concavity does not certify an arbitrary numerical routine: simplex feasibility and inner/dual residuals still require measurement. The empirical tensor itself remains an approximation to population preferences; exponentiation, normalization, and log-sum-exp do not preserve unbiasedness of noisy margins.

GPT: R34 — Quantify the approximation introduced before binary regression even starts. An unbiased estimate of m does not give an unbiased opponent after exponentiation and normalization, nor an unbiased log-sum-exp game-value estimate. The practical target therefore refers to a random finite-pool opponent, not automatically to the population Q_k^π in the proof. With $L = 4$ and four reference candidates, this discrepancy may be important, especially at small β and near zero surplus. On a finite game with known preference probabilities, vary L , comparator count, and temperature; measure opponent-distribution error, target bias/variance, surplus bias, and final policy error against exact quantities. State whether labels used to select/reweight opponents are reused in the target; use fresh labels or cross-fitting to diagnose selection effects. Separate this estimation error from neural regression error and numerical dual accuracy.

Response: R34 — Partially addressed. The implementation now defines a finite-pool teacher explicitly and acknowledges that exponentiation, normalization, and log-sum-exp do not preserve unbiasedness of noisy margins. It separates numerical solver accuracy from sampling and neural errors, but the requested sampling-count, comparator-count, temperature, and label-reuse or cross-fitting error audit remains pending.

Finite-pool feasibility and failure reporting. When diagnosing an uncertified solve, use the finite-pool program

$$\hat{\tau}^* = \max_{\{p_x\}, \tau} \tau \quad \text{s.t.} \quad \mathbb{E}_{x \sim \hat{\mathcal{D}}} \hat{V}_{k,x}(p_x) - \hat{d}_k \geq \tau \quad (\forall k), \quad p_x \in \Delta^{n_x}.$$

This is convex optimization with concave-superlevel constraints, not generally a linear program at $\beta_k > 0$. A verified positive feasible margin witnesses strict finite-pool improvement. Only a valid nonpositive upper bound certifies its absence; failure to find a witness is unresolved. A tolerance-bounded interval near zero means no resolved positive margin, not exact infeasibility. Independent learner and reference banks need not contain a categorical representation of the fallback, so finite-pool infeasibility is not population infeasibility.

For an analytic infeasible example, take two actions, $\mu = (1/2, 1/2)$, $A_1 = \begin{pmatrix} 0 & 1/2 \\ -1/2 & 0 \end{pmatrix}$ and $A_2 = -A_1$. For $p = (a, 1 - a)$ and any common $\beta > 0$, $s_1(p) = (a - 1/2)/2 = -s_2(p)$, hence $\tau^* = 0$. No clipping or inverse-surplus floor creates a strictly feasible Nash target.

The execution contract records feasible, certified-infeasible, numerical-unresolved, and neural-rejected outcomes separately. Uncertified solves retain the previous policy. This audit introduces no separate neural warm-start stage. Post-fit acceptance uses prespecified development checks, including empirical surplus signs, target fit, and numerical validity; reused monitoring prompts are validation data. False acceptance/rejection rates near the boundary remain an unmeasured audit, and generic bounded-mean confidence formulas do not automatically cover biased game-value estimates.

A.6 LAST-ITERATE CONVERGENCE

Theorem 1 (Last-iterate convergence). *Under Assumption 1, let π^* maximize (12), set $\pi_0 = \pi_{\text{ref}}$, and solve (16) exactly at every outer iteration with arbitrary $\eta_t > 0$. Then $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and for every $T \geq 1$,*

$$F(\pi^*) - F(\pi_T) \leq \frac{D(\pi^* \| \pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (31)$$

Work throughout in the $\mathcal{D}$ -weighted geometry: pair functions of (x, y) by

$$\langle f, g \rangle_{\mathcal{D}} = \mathbb{E}_{x \sim \mathcal{D}} \sum_y f(x, y) g(x, y),$$

and let h be the *prompt-wise* negative entropy $h(\pi)(x) = \sum_y \pi(y | x) \log \pi(y | x)$, whose gradient at (x, y) is $1 + \log \pi(y | x)$. Its Bregman divergence in that geometry is exactly $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$. The prompt expectation enters once, through the pairing, and is not also carried inside h . Because the objective is increasing in u , the policy part of (16) is equivalently

$$\pi_{t+1} \in \arg \max_{\pi \in \Pi_+} \left\{ F(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (32)$$

The feasible set is nonempty by Assumption 1, and $F(\pi) \rightarrow -\infty$ as any surplus approaches zero. Thus the maximizer exists in Π_+ even when the center π_t is the reference $\pi_{\text{ref}} \notin \Pi_+$. Because π_t has full support and the remaining objective has finite directional derivatives at an interior surplus point, the inward derivative of $-D(\pi\|\pi_t)$ excludes zero-probability coordinates. Induction therefore gives a full-support, positive-surplus π_{t+1} .

Let g_{t+1} be a supergradient of F at π_{t+1} . First-order optimality of (32) and concavity of F imply, for every $\pi \in \Pi_+$,

$$F(\pi) - F(\pi_{t+1}) \leq \langle g_{t+1}, \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (33)$$

$$\leq \frac{1}{\eta_t} \langle \nabla h(\pi_{t+1}) - \nabla h(\pi_t), \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (34)$$

$$= \frac{1}{\eta_t} [D(\pi\|\pi_t) - D(\pi\|\pi_{t+1}) - D(\pi_{t+1}\|\pi_t)]. \quad (35)$$

For $t \geq 1$, π_t is feasible in the next subproblem. Comparing the optimum with π_t gives

$$F(\pi_{t+1}) - \frac{1}{\eta_t} D(\pi_{t+1}\|\pi_t) \geq F(\pi_t), \quad (36)$$

so the objective values are nondecreasing from π_1 onward.

Set $\pi = \pi^*$ in (35), multiply by η_t , and sum from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \eta_t (F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^*\|\pi_{\text{ref}}). \quad (37)$$

Because the gaps are nonincreasing, the left-hand side is at least $(\sum_t \eta_t)(F(\pi^*) - F(\pi_T))$, which proves (31).

Corollary 1 (Convergence of the surplus profile). *Under the conditions of Theorem 1, let $s^* = s(\pi^*)$ denote the unique surplus vector induced by a Nash-optimal policy. Then*

$$\|s(\pi_T) - s^*\|_2^2 \leq \frac{2D(\pi^*\|\pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (38)$$

Hence a constant proximal step gives $\|s(\pi_T) - s^*\|_2 = O(T^{-1/2})$.

Proof. This is a stronger consequence of Nash-welfare convergence rather than a separate requirement for the NBPO update. For every objective, $V_k(\pi) \in [-1/2, 1/2]$, and $d_k = V_k(\pi_{\text{ref}}) \leq 0$ because $\nu = \pi_{\text{ref}}$ is feasible and $J_k(\pi_{\text{ref}}, \pi_{\text{ref}}) = 0$. Hence every positive surplus is at most one. On $(0, 1]^K$, the function $G(s) = \sum_k \log s_k$ is 1-strongly concave in Euclidean norm. Proposition 2 implies that all Nash-optimal policies induce the same game-value vector, and therefore the same surplus vector s^* . First-order optimality over the convex comprehensive surplus set induced by (14) gives

$$G(s^*) - G(s) \geq \frac{1}{2} \|s - s^*\|_2^2. \quad (39)$$

Applying this inequality to $s = s(\pi_T)$ and using (31) proves (38). $\square$

B CONTROLLED BENCHMARK: CONSTRUCTION AND FULL RESULTS

Figure 2 in the main text summarizes this benchmark. The construction, every method, and robustness checks are retained below.

Feasibility-preserving construction. For every seed we form a transitive base matrix and an odd-cardinality circulant skew-symmetric perturbation. Zero row sums keep the uniform reference disagreement invariant, while bisection selects the cycle magnitude so that $\rho^* = \max_{\pi} \min_k s_k(\pi)$ stays close to a target margin. We reject invalid probabilities rather than clip matrix entries. The resulting panel increases BT deviance from approximately 2.7×10^{-4} to 0.351 while retaining positive joint improvement in every instance. All policies are evaluated under the same adaptive-game values, not their method-native scales.

Table 5: Full controlled results over five seeds under the common adaptive-game evaluator. Residuals are method-specific; the legacy map is a non-convergence ablation.

method	min surplus	sd	min s/ρ^*	sd exploitability	inner resid.
$\alpha = 0.00, \rho^* = 0.0300, BT \text{ dev.} = 0.0003$					
NBPO direct	+0.0264	0.0011	0.881	0.037	2×10^{-12}
Game-KS	+0.0268	0.0005	0.893	0.017	10^{-9}
Game-util.	+0.0260	0.0014	0.866	0.048	3×10^{-10}
Fixed-ref. Nash	+0.0265	0.0011	0.883	0.036	0
BT-RM-Nash	+0.0265	0.0011	0.883	0.036	0
BT-RM-util.	+0.0260	0.0014	0.867	0.047	0
legacy R -step [†]	+0.0252	0.0012	0.840	0.041	5×10^{-10}
$\alpha = 0.25, \rho^* = 0.0300, BT \text{ dev.} = 0.0202$					
NBPO direct	+0.0269	0.0013	0.897	0.044	10^{-7}
Game-KS	+0.0272	0.0007	0.905	0.025	10^{-7}
Game-util.	+0.0265	0.0017	0.883	0.058	4×10^{-8}
Fixed-ref. Nash	+0.0272	0.0012	0.907	0.040	0
BT-RM-Nash	+0.0272	0.0012	0.907	0.040	0
BT-RM-util.	+0.0269	0.0015	0.895	0.049	0
legacy R -step [†]	+0.0258	0.0014	0.860	0.046	7×10^{-1}
$\alpha = 0.50, \rho^* = 0.0300, BT \text{ dev.} = 0.0813$					
NBPO direct	+0.0274	0.0013	0.914	0.044	2×10^{-6}
Game-KS	+0.0276	0.0007	0.921	0.023	2×10^{-7}
Game-util.	+0.0270	0.0017	0.900	0.057	2×10^{-7}
Fixed-ref. Nash	+0.0243	0.0018	0.809	0.059	0
BT-RM-Nash	+0.0243	0.0018	0.809	0.058	0
BT-RM-util.	+0.0240	0.0019	0.800	0.064	0
legacy R -step [†]	+0.0025	0.0044	0.082	0.147	10×10^{-1}
$\alpha = 0.75, \rho^* = 0.0300, BT \text{ dev.} = 0.1882$					
NBPO direct	+0.0278	0.0013	0.926	0.042	8×10^{-6}
Game-KS	+0.0278	0.0005	0.928	0.018	8×10^{-7}
Game-util.	+0.0273	0.0017	0.909	0.056	7×10^{-7}
Fixed-ref. Nash	+0.0138	0.0027	0.460	0.089	0
BT-RM-Nash	+0.0138	0.0027	0.460	0.089	0
BT-RM-util.	+0.0137	0.0028	0.455	0.095	0
legacy R -step [†]	-0.0453	0.0060	-1.511	0.201	10×10^{-1}
$\alpha = 1.00, \rho^* = 0.0283, BT \text{ dev.} = 0.3510$					
NBPO direct	+0.0265	0.0033	0.931	0.041	5×10^{-5}
Game-KS	+0.0264	0.0027	0.932	0.019	3×10^{-6}
Game-util.	+0.0258	0.0037	0.908	0.063	2×10^{-6}
Fixed-ref. Nash	-0.0063	0.0068	-0.241	0.277	0
BT-RM-Nash	-0.0063	0.0068	-0.242	0.276	0
BT-RM-util.	-0.0065	0.0070	-0.248	0.284	0
legacy R -step [†]	-0.0949	0.0064	-3.373	0.406	10^0

Table 6: Finite-pool solver audit at $\alpha = 1$ in the original controlled family. The direct practical solve recovers the positive-surplus proximal target; the deployed legacy map does not.

solution	min surplus	$/\rho^*$	inner residual	TV to exact prox.
Exact global Nash	+0.0158	+0.935	—	0.007
Exact proximal Nash	+0.0158	+0.936	—	0.000
NBPO direct inner solve	+0.0139	+0.821	2×10^{-6}	0.027
Legacy R -step map	-0.1851	-10.938	10^0	0.643
BT-RM-Nash	-0.1001	-5.918	0	0.495

Robustness across construction and temperature. The primary construction is checked against an independently generated projected skew-symmetric family, $C = N(R - R^\top)N$ with N the symmetric projector annihilating both μ and a witness policy, and against a temperature sweep on that family. The trend replicates: adaptive-game rows stay flat while fixed-reference and scalar-reward rows fall. The effect *size* does not: the $\alpha = 1$ gap is +0.111 on the projected family against +1.172 on the circulant one, because the projected family’s BT deviance only reaches 0.174 against 0.351. We therefore claim robustness of the trend and do not attach the circulant effect size to it. Intervals in Table 7 are exact paired Student- t intervals over seeds, not bootstrap intervals.

Table 7: Robustness across construction and temperature. Gaps in normalized worst surplus use paired Student- t 95% intervals across seeds. The temperature sweep uses the projected family.

cycle family	β	seeds	NBPO—FixedRef (t_{95})		NBPO—BT-RM (t_{95})		min ρ^*	converged
Circulant (primary)	0.25	5	+1.1724	[+0.8715, +1.4733]	+1.1733	[+0.8735, +1.4731]	0.0283	150/150
Projected skew-symmetric	0.25	5	+0.1114	[+0.0714, +0.1514]	+0.1115	[+0.0715, +0.1514]	0.0332	150/150
Projected skew-symmetric	0.10	3	+0.2381	[+0.0880, +0.3883]	+0.2385	[+0.0883, +0.3887]	0.0331	90/90
Projected skew-symmetric	0.50	3	+0.0448	[+0.0165, +0.0730]	+0.0449	[+0.0167, +0.0730]	0.0334	90/90

Table 8: Released-supervision audit. Zero observed human triangles means cyclicity is unidentifiable, not absent. Human conflict is cross-objective label disagreement on held-out pairs.

dataset	rows	prompts	responses	obs. triangles	human conflict	supervision	role
SAFERLHF	73,907	38,639	143,668	0	24.33%	direct pairwise	objective conflict
ULTRAFEEDBACK	63,967	63,967	4/prompt	n/a (score-induced)	n/a	ordinal attributes	transitive control

C NATURAL SUPERVISION AND PREFERENCE MODELS

Dataset audit. Table 8 records the empirical role of each released dataset. ULTRAFEEDBACK supplies complete per-completion ordinal attributes, so score-induced pairwise labels are transitive by construction. SAFERLHF supplies direct helpfulness and safety comparisons but almost exclusively isolated edges; zero observed triangles means the data cannot test within-objective cycles. It nevertheless contains substantial cross-objective disagreement and is therefore appropriate for the bargaining experiment.

C.1 WITHIN-OBJECTIVE CYCLIC-PREFERENCE AUDIT

GPT: R32 — Make the planned cycle-conditioned analysis distinguish misspecification from estimation noise. Define the cyclicity statistic and subset threshold before examining which method wins. Report strict majority cycles separately from ties, presentation-order disagreement, and unstable edges. Repeat a subset of judgments under a specified randomness source: rerunning an identical deterministic judge prompt does not create independent evidence. Fit a BT projection to the same comparison matrices and compare original-matrix versus projected-matrix bargaining under matched budgets. Combine this with a synthetic sweep of cycle strength/frequency at finite β and full-support references, while preserving joint feasibility. These controls test whether nontransitivity itself explains adaptive-opponent performance; merely stratifying noisy four-response matrices can instead select difficult prompts or judgment variance.

Response: R32 — Partially addressed. The completed 60,000-judgment audit reports repeated cycles, coverage, order sensitivity, fitted BT-noise comparisons, and held-out prediction. No repeated-cycle excess is statistically resolved; no order-robust witness is eligible, so no additional confirmation judgments are collected. Original-versus-BT-projected bargaining remains pending, and the null findings are not interpreted as proof of transitivity.

Across 100 reserved prompts and 60,000 judgments, no rubric shows a repeated-cycle excess over the fitted BT noise control. No triangle satisfies the order-robust witness rule, so no confirmation judgments were collected. These results neither establish transitivity nor demonstrate cyclicity in human feedback. We test whether directly elicited preferences remain cyclic within each training rubric. PROSPER reports residual cycles when LLM judges evaluate checklist items separately (Zhang et al., 2026); we adapt this comparison to our fixed objectives and add independent repeat judgments and a fitted sampling-noise control. This audit concerns new pairwise feedback on natural prompts, not cycles in UltraFeedback’s score-induced labels.

Prompts and responses. Reserve 100 eligible UltraFeedback prompts, stratified across Ultra-Chat, ShareGPT, and Evol-Instruct, from unused prompt groups disjoint from all five frozen training/development/final splits and the benchmark bank. Use a separate 20-prompt pilot for parser and throughput checks; exclude it from the audit. Freeze prompt IDs, rubrics, model revisions, seeds, and this protocol before viewing judgments. Generate four responses per audit prompt from the unchanged Llama-3.1-8B-Instruct reference, with temperature 0.8, top- $p = 0.9$, and at most 2048

new tokens. Judge all six unordered pairs. Retain duplicates and truncated generations, flag both, and compare the full recorded texts without preference-based filtering. Thus the audit targets this declared candidate distribution. Extra generators or larger pools require separately labeled results.

Objectives and repeated judgments. Use the same written IF, truthfulness, honesty, and helpfulness rubrics as the UF-4 experiment, one rubric per query. A fifth, joint-rubric condition asks for an overall comparison and serves only as an aggregation control. Run the local Qwen3-14B judge with thinking disabled, temperature 0.8, top- $p = 0.9$, top- $k = 20$, and a 256-token output limit. Unlike a normalized Likert score, the primary response is an explicit choice: A, B, or tie. Require valid structured output, hide model identities, and include the complete prompt and both responses within a declared 8192-token input budget. Out-of-budget inputs are reported as missing, not silently shortened.

Collect two independent panels, A and B, each with five judgments per presentation order. Use fresh seeds and isolated contexts; correct the response orientation before analysis. The core budget is $100 \times 6 \times 5 \times 20 = 60,000$ judgments, excluding the pilot and confirmation below. Report throughput and GPU-hours. Invalid outputs remain missing without selective retries; a complete prompt-rubric tournament requires all scheduled judgments on all six pairs. Report retained counts and missingness by cause.

Preference graphs and metrics. For each correctly oriented judgment, assign $X = 1$ to a win, $X = 0$ to a loss, and $X = 1/2$ to a tie. For panel $b \in \{A, B\}$, define

$$\hat{p}_{ij}^{(b)} = \frac{1}{2} \left(\bar{X}_{ij}^{(b, i\text{-first})} + \bar{X}_{ij}^{(b, j\text{-first})} \right). \quad (40)$$

This is a tie-adjusted preference under equal presentation-order sampling, conditional on the fixed judge, rather than a human-population probability. An edge estimate requires all ten scheduled judgments; otherwise it is unresolved. Draw $i \rightarrow j$ only when $\hat{p}_{ij}^{(b)} > 1/2$; exact ties create no edge. Let C_A count prompts containing any directed cycle in panel A, including length-four cycles, and let R count those containing the *same oriented response triangle* in both panels. The latter measures repeatability, not statistical confirmation. Let W_A count prompts where every response has an observed loss; this rules out a strict winner in the observed graph. Otherwise, a graph without an observed strict winner is unresolved. Report all three rates over the 100 planned prompts, and cyclic-triangle rates over the 400 planned triangles. Missing comparisons cannot establish an event; report unresolved counts and the corresponding worst-case upper rate, without interpreting unobserved cycles as absent. Let n be the number of tournaments complete in both panels.

BT sampling-noise control. A noisy transitive model can also produce observed cycles. Fit an order-aware BT model with a tie extension to panel A, with one score r_i per response, a rubric-level position coefficient b , and tie parameter $\nu > 0$:

$$\Pr(i, \text{tie}, j \mid o) \propto \left(e^{z/2}, 2\nu, e^{-z/2} \right), \quad z = r_i - r_j + bo, \quad o \in \{-1, +1\}. \quad (41)$$

Here $o = +1$ presents i first; response scores sum to zero within each prompt. Fit by summed negative log likelihood with a fixed 10^{-3} squared penalty on scores, b , and $\log \nu$ to keep separated fits finite. Under equal-order averaging this null retains a transitive latent ranking. Generate 2000 synthetic A/B panel pairs under the fitted null, preserving presentation orders, repeat counts, and missingness, and rerun the graph construction. Report its mean repeated-cycle rate R_{BT} , the excess $\Delta R = R - R_{\text{BT}}$, and the upper-tail value $(1 + \#\{R^{\text{sim}} \geq R\})/2001$. These are approximate fitted-null checks, not exact composite-null tests. Apply Holm correction across the four individual rubrics; report the joint condition separately. Obtain 95% intervals with 2000 whole-prompt bootstrap replicates, retaining all judgments and refitting the null. These intervals measure prompt variability, not all uncertainty in the fitted noise model. Pairs and triangles are not independent samples.

Coverage and training relevance. Table 10 records coverage, ties, order sensitivity, and whether the frozen GPM predicts the new comparisons better than the frozen scalar BT model. Score both on panel B using the mean per-judgment Brier loss $(P - X)^2$ with paired prompt-bootstrap intervals; neither model is fitted or calibrated on the audit. This predictive comparison does not itself establish cycles. Label any model subsequently trained on the new judgments as a separate experiment. A

Table 9: Cycle audit on 100 prompts. n counts tournaments complete in both panels; ΔR is the repeated-cycle excess in percentage points over the fitted BT null, with 95% prompt-bootstrap intervals. All four Holm-adjusted checks are nonsignificant.

Rubric	n	C_A	R	R_{BT}	W_A	ΔR [CI]	p_{Holm}
IF	77	9	1	0.97	7	+0.03 [−2.00, 1.00]	1.00
Truthfulness	73	2	0	0.43	2	−0.43 [−2.00, 0.00]	1.00
Honesty	75	5	1	0.63	4	+0.37 [−2.00, 1.00]	1.00
Helpfulness	96	4	0	0.31	2	−0.31 [−2.00, 0.00]	1.00
Joint control	90	1	0	0.24	1	−0.24 [−1.00, 0.00]	—

C_A : any cycle in A; R : the same oriented triangle in A and B; W_A : every response has an observed loss in A. Both-panel incomplete counts are 23/27/25/4/10 in row order. Unresolved cases are not cycle-free; the joint control is outside the four-rubric Holm family.

planned SafeRLHF extension would repeat the audit under helpfulness and harmlessness; a claim about human preferences additionally requires fresh human comparisons on complete response triples under those same rubrics. Human annotations, prompted-judge choices, and teacher predictions must remain separate evidence sources.

Table 10: Audit coverage and prediction. Order gap measures presentation-order sensitivity; negative GPM−BT Brier differences favor GPM. All paired 95% intervals include zero; neither model was fitted or calibrated on this audit.

Rubric	Missing (%)	Ties (%)	Triangles (%)	Order gap	BT Brier	GPM−BT [CI]
IF	0.60	20.8	3.25	0.535	0.2077	−0.0016 [−0.0075, +0.0046]
Truthfulness	0.48	64.9	0.50	0.125	0.0999	−0.0038 [−0.0087, +0.0012]
Honesty	0.36	29.3	1.25	0.305	0.1933	−0.0029 [−0.0100, +0.0037]
Helpfulness	0.12	6.9	1.00	0.695	0.2341	−0.0030 [−0.0079, +0.0014]
Joint control	0.41	3.9	0.25	0.692	n/a	n/a

Missing: fraction of scheduled judgments unavailable; ties: fraction of valid judgments labeled tie; triangles: observed cyclic triangles in A divided by 400; order gap: mean absolute difference between the two correctly oriented order-specific means in A, on complete pairs. Also release unresolved counts, missingness causes, and duplicate/truncation counts. Brier comparisons use identical complete panel-B pairs.

Independent confirmation of a cycle. Before opening B, select at most one directed cyclic triangle per individual rubric from A. Require complete A judgments and all three cycle directions to agree across both presentation orders. Among eligible cycles, maximize the minimum edge margin, breaking ties by prompt and response IDs. Fix its orientation and allocate 200 independent B judgments per selected edge, 100 per order. This adds at most $4 \times 3 \times 190 = 2280$ judgments; only the first five B judgments per presentation order enter the aggregate tables and their null simulations. For each selected edge e , use the full $m = 200$ bounded observations to test its order-averaged population mean against $1/2$:

$$p_e = \exp \left[-2m (\max\{\hat{p}_e - 1/2, 0\})^2 \right], \quad p_{\text{cycle}} = \max_{e \text{ in triangle}} p_e. \quad (42)$$

Hoeffding’s bound gives a conservative edge test under independent judgments (Hoeffding, 1963). The cycle null states that at least one of its three edges has mean at most $1/2$, so the maximum edge p -value tests that union null. Apply Holm at 0.05 across the four rubric witnesses, assigning $p = 1$ when no witness is selected or confirmation is incomplete. This establishes a cycle only for the specified judge’s order-averaged preference distribution. Require no adaptively chosen extra samples after viewing B. Report both order-specific means, and release the prompt, rubric, response texts, and individual judgments.

Interpretation. The primary witness tests select no eligible triangle. Across the four individual rubrics, panel A contains 24 cyclic triangles, but only 15/72 constituent edges retain their direction in both presentation orders. This fails the witness rule without proving that order-averaged cycles are artifacts. Secondary fitted-null checks detect no repeated-cycle excess, and the frozen GPM−BT Brier intervals all include zero. These outcomes do not establish transitivity, human-preference cyclicity, or an advantage of Nash aggregation.

Table 11: Cycle-confirmation outcome. No individual rubric produced an eligible order-robust witness; no additional judgments were collected, and the protocol assigns $p = 1$. This does not establish transitivity.

Rubric	Triangle ID	m	$\hat{p}_{ab}$	$\hat{p}_{bc}$	$\hat{p}_{ca}$	p_{Holm}	Confirmed?
IF	none selected	—	—	—	—	1.00	Not confirmed
Truthfulness	none selected	—	—	—	—	1.00	Not confirmed
Honesty	none selected	—	—	—	—	1.00	Not confirmed
Helpfulness	none selected	—	—	—	—	1.00	Not confirmed

Triangle ID resolves the prompt and response IDs (a, b, c). Add the full prompt, rubric, and three response texts for each selected case. A missing candidate is reported as “none selected”; it is never replaced using B.

C.2 PREFERENCE-MODEL DIAGNOSTICS

Anti-symmetric general preference model. For objective k , a joint encoder produces $a_{\phi,k}(x, y, z)$ and we set

$$\ell_{\phi,k}(x, y, z) = \frac{1}{2}\{a_{\phi,k}(x, y, z) - a_{\phi,k}(x, z, y)\}, \quad P_{\phi,k}(y \succ z \mid x) = \sigma(\ell_{\phi,k}(x, y, z)).$$

This guarantees exact swap complementarity and self-ties without constraining the logit to a scalar reward difference. In the SafeRLHF diagnostic, both the joint model and the scalar BT baseline use RoBERTa-base so that encoder capacity and training cost are matched; this backbone choice is an experimental control rather than part of the NBPO method. Because the observed SafeRLHF graph contains no triangles, their near-null comparison is a calibration result rather than evidence that human preferences are transitive.

C.3 GENERAL-INSTRUCTION DATA AND PREFERENCE MODELS

The UltraFeedback panel retains UltraChat, ShareGPT, and Evol-Instruct, excludes TruthfulQA, FalseQA, and FLAN, and removes benchmark overlaps before splitting by normalized prompt identity. The realized split is 20k/2k preference-model train/dev, 10k/1k policy train/dev, and 2k final evaluation; final prompts are excluded from model selection. Filtering retained 38,240 eligible groups (Table 14), so the predeclared 5k policy-training fallback for 30k–35k eligible groups, excluding 35k, was not used; fewer than 30k would require revising the budget. Honesty was the binding attribute: 1,494 groups lacked two valid ratings, versus at most one group for each other attribute. Split hashes are recorded in the experiment artifacts.

Released GPT-4 ratings train four attribute heads: valid comparisons receive 0/1 labels, ties receive 0.5, and missing labels are masked. Four completions yield six unordered pairs per prompt. These rating-derived labels are transitive by construction and cannot establish preference cycles. Policy-training prompts are disjoint and use newly sampled base-policy responses.

The GPM and scalar BT model share the 149M ModernBERT-base encoder with native 8192-token support (Warner et al., 2024), supervision, prompt splits, and one preference-model seed. The GPM uses the antisymmetric construction above; BT uses differences of four scalar scores. Training uses AdamW, learning rate 2×10^{-5} , batch 32 pairs, weight decay 0.01, warmup 0.06, length bucketing, and up to two epochs, selected on preference-model development data. Policy-seed repeats therefore do not measure teacher uncertainty. Unsupported inputs are audited; an independent judge evaluates full generated responses.

Policy training uses full 8B fine-tuning, batch 32 pairs, the existing peak learning-rate scale, and 1250 updates for the realized 10k training prompts; the unused 5k fallback specifies 625 updates. This gives four sampled pair exposures per prompt on average. Each run has its own cosine schedule and one outer stage. Independent learner and comparator pools contain eight occurrences each; reference–reference scores and the shared tensor are cached across methods. Two encoder orders and four joint heads require $2 \times 10^4(64 + 28) = 1.84$ million encoder passes for 10k prompts.

The 20.00% generation-cap rate counts responses reaching the 1024-token horizon; the 0/500,000 scoring-oversize count concerns serialized encoder inputs. These have different denominators. Reference-tensor antisymmetry and zero diagonal are imposed by construction and do not validate the teacher.

Table 12: Held-out validation of checkpointed GPM and BT ensembles, seeds 41/42/43. Calibration uses validation data. Sparse comparison graphs cannot establish a cyclicity advantage.

model	objective	per-seed (mean of 3)			T	calibrated 3-seed ensemble				
		NLL	bal. acc	AUC		NLL	bal. acc	AUC	ECE	ens. sd
Anti-symmetric GPM	helpfulness	0.5543	0.7218	0.7999	1.282	0.5512	0.7257	0.8034	0.1228	0.0293
Anti-symmetric GPM	harmless.	0.5180	0.7556	0.8464	1.297	0.5131	0.7573	0.8512	0.1782	0.0357
Scalar BT	helpfulness	0.5560	0.7214	0.7974	1.270	0.5527	0.7250	0.8009	0.1220	0.0300
Scalar BT	harmless.	0.5212	0.7507	0.8431	1.265	0.5163	0.7544	0.8484	0.1779	0.0357

Table 13: Generated-pool pilot on 200 development prompts. Conflict and cycles are model predictions, not human-label measurements; cycles use a common comparator tournament.

geom.	dup. rate	entropy	ens. sd	GPM/BT sign	model conflict	pred. cycles	split-half r	ρ	sign agr.	dual evals	solve (s)
4+4	0.0073	0.6117	0.0429	0.9102	7.67%	0/800	0.9172	0.9132	0.8667	42	24.8
8+8	0.0167	0.6120	0.0414	0.9061	7.80%	10/11200	0.9242	0.9180	0.8925	30	18.4

Table 15 reports neural fitting diagnostics. For comparison, SafeRLHF achieved $\text{nMSE } .6845 \pm .0048$ and sign accuracy $.7009 \pm .0016$ over three seeds, with target RMS 1.125; normalized errors should be interpreted alongside these different target scales. Nash and utilitarian targets have total variation .0195, and the two arms attain similar fitting accuracy. Neither observation establishes a policy-quality difference.

The pilot showed substantial first-position skew (Table 17); the audit evaluates both presentation orders and reports their average. Capability measurements in Table 16 are separate from the four alignment objectives.

Aggregation across objectives and across prompts. NBPO applies $\sum_k \log(V_k(\pi) - d_k)$ to prompt-averaged objective values. In contrast, PROSPER’s regularized criterion has prompt-dependent weights (Zhang et al., 2026):

$$\min_{w: \mathcal{X} \rightarrow \Delta_K, \nu} \mathbb{E}_x \left[\sum_k w_k(x) J_{k,x}(\pi, \nu) + \beta \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)) \right],$$

where $J_{k,x}$ is the conditional pairwise payoff. Constant shifts from centered payoffs do not change its optimizer. Prompt-dependent weights cannot be replaced by a shared weight without changing this criterion. Prompt-direct optimization decomposes only the inner computation; NBPO’s λ remains shared across prompts.

To illustrate aggregation order alone, take two equally likely prompts. A policy with surplus vectors $(.2, 0)$ and $(0, .2)$ has average $(.1, .1)$; another with $(.08, .08)$ on both prompts has average $(.08, .08)$. Global Nash prefers the first, whereas the average prompt-wise minimum is 0 versus .08 and prefers the second. This schematic example omits opponent regularization and is not a numerical PROSPER result. We therefore distinguish a faithful PROSPER adaptation from global game-maxmin over NBPO values.

D TRAINING PROTOCOL AND BASELINES

GPT: R28 — Diagnose the failure cases already present in the objective-wise table. In Table [original-draft ref.: tab:objectivewise-appendix], TL;DR conciseness has NBPO surplus $-.001$. On SafeRLHF, NBPO’s vector $(.118, .101)$ is strictly dominated in both reported coordinates by PROSPER $(.151, .116)$, utilitarian aggregation $(.157, .138)$, and both maxmin controls. These estimates do not refute the exact population theorem: finite proximal regularization, estimation, and neural error can all prevent its limiting outcome properties from appearing. They do prevent interpreting this trained policy as empirically individually rational or Pareto efficient on these diagnostics. Verify the comparisons under identical prompts, evaluator, comparators, and temperature with paired uncertainty. Then localize the gap to finite-pool optimization or neural realization, and discuss the verified failure cases in the main results rather than only the appendix.

Table 14: Reported UF-4 data, preference-model, pool, and solver measurements. IF, Truth, Honesty, and Help denote instruction-following, truthfulness, honesty, and helpfulness.

Stage	Quantity	Value
Data	Raw rows: UltraChat / ShareGPT / Evol-Instruct	9,929/19,949/10,000
	Exact normalized prompts	39,846
	Groups after verified near-duplicate merge	39,743
	Benchmark-overlap groups removed (all AlpacaEval)	9
	Groups overlapping SafeRLHF	0
	Eligible groups: two rated completions per objective	38,240
	Realized regime	target sizes, no fallback
	Eligible groups left unassigned	3,240
	Groups with fewer than two honesty ratings	1,494
	Preference-model pairs, train / dev	120000/12000
Supervision	Tie fraction, IF / Truth / Honesty / Help	0.253/0.481/0.308/0.262
	Masked train pairs (honesty only)	3,668
Teacher	GPM selected step / mean dev NLL / worst NLL	3750/0.5601/0.6305
	GPM dev accuracy, non-ties: IF / Truth / Honesty / Help	0.7958/0.7469/0.7677/0.8346
	BT selected step / mean dev NLL / worst NLL	3500/0.5589/0.6283
	BT dev accuracy, non-ties: IF / Truth / Honesty / Help	0.7995/0.7527/0.7599/0.8403
	Oversize / total serialized training pair sequences	216/240,000
Pool	Response truncations / maximum serialized tokens	0/4,323
	Sampled occurrences (learner 8 + comparator 8)	160,000
	Responses stopped by the 1024-token cap	20.00%
Scoring	Duplicate occurrences retained with multiplicity	725
	GPM encoder examples, realized / predicted	1,840,000/1,840,000
	Aggregate throughput on four H200s	577.6 sequences/s
	Oversize / total serialized scoring sequences	0/500,000
Solve	Reference-tensor skew / diagonal residual	0.0/0.0
	Nash λ , IF / Truth / Honesty / Help	7.4627/11.0426/8.1138/7.3043
	Nash λ L_1	33.9235
	Nash train surplus, IF / Truth / Honesty / Help	0.1340/0.0906/0.1232/0.1369
	Independently computed Nash dual residual, train	3.74×10^{-14}
	Maximum inner stationarity spread over prompts	1.88×10^{-10}

Table 15: UF-4 neural projection diagnostics on held-out targets. These measure target realization, not the quality of generated responses; three-seed summaries use sample standard deviation.

Run	Quantity	Value
NBPO-MSE, seed 42	Optimizer updates, completed / planned	1,250/1,250
	Held-out dev pairs per evaluation	28,000
	Dev nMSE $\downarrow$, first $\rightarrow$ last of five evaluations	0.6083 $\rightarrow$ 0.5823
	Dev sign accuracy $\uparrow$, first $\rightarrow$ last	0.6605 $\rightarrow$ 0.6694
	Final dev Pearson / Spearman	0.6515/0.5479
	Dev target RMS / fitted log-ratio RMS	5.4711/3.8409
	Wall time, four H200s / GPU-hours	2.79 h / 11.2 GPU-h
	Optimizer updates, completed / planned	1,250/1,250
	Dev nMSE $\downarrow$, first $\rightarrow$ last of five evaluations	0.6061 $\rightarrow$ 0.5862
	Dev sign accuracy $\uparrow$, first $\rightarrow$ last	0.6603 $\rightarrow$ 0.6702
Utilitarian, seed 42	Final dev Pearson / Spearman	0.6488/0.5480
	Dev target second moment, Nash / utilitarian	29.9332/31.2487
	Final dev nMSE $\downarrow$, mean $\pm$ SD	0.5861 $\pm$ 0.0033
NBPO-MSE, three seeds	Final dev sign accuracy $\uparrow$, mean $\pm$ SD	0.6698 $\pm$ 0.0017
	Final dev Pearson, mean $\pm$ SD	0.6490 $\pm$ 0.0023
	Final nMSE, seeds 42 / 43 / 44	0.5823/0.5881/0.5878
Utilitarian, three seeds	Final dev nMSE $\downarrow$, mean $\pm$ SD	0.5800 $\pm$ 0.0054
	Final dev sign accuracy $\uparrow$, mean $\pm$ SD	0.6732 $\pm$ 0.0059
	Final nMSE, seeds 42 / 43 / 44	0.5862/0.5778/0.5761
Fixed-reference Nash	Train λ (IF / Truth / Hon / Help)	7.6311/11.1646/8.2602/7.4579
	Train λ L_1 , fixed-reference / adaptive	34.5138/33.9235
	Train minimum surplus, fixed-reference / adaptive	0.08957/0.09056
	Dual box active	no

Response: R28 — Partially addressed. The main text retains the unresolved Nash–utilitarian comparison and the limits of teacher-defined gains after moving detailed fit tables to the appendix. The legacy TL;DR violation and SafeRLHF dominance comparisons still require source/protocol reconciliation and paired evaluation; moving tables does not resolve these failures.

Table 16: UF-4 capability measurements under the shared evaluation recipe. These point estimates do not establish equivalence; trained-policy IFEval and safety measurements remain pending.

Checkpoint(s)	Benchmark / metric	Value
Base	ARC-Challenge 25-shot, acc / acc_norm	0.5674/0.6049
	HellaSwag 10-shot, acc / acc_norm	0.5927/0.8008
	IFEval strict-prompt / strict-instruction	0.7542/0.8249
	GSM8K exact match	0.8666
	HarmBench harmful-completion rate	0.2813
Base / NBPO / utilitarian	XSTest 450	unadjudicated
	ARC-Challenge 25-shot, acc_norm	0.6049/0.6058/0.6032
	HellaSwag 10-shot, acc_norm	0.8008/0.7984/0.7978
Base / NBPO	MMLU 5-shot, acc (utilitarian pending)	0.6823/0.6816

Table 17: UF-4 audit execution diagnostics. The 20 pilot prompts are excluded from audit estimates; no candidate cycle met the confirmation eligibility rule.

Stage	Quantity	Value
Pilot	Judgments collected / planned	12,000/12,000
	Explicit-verdict parse rate	0.9959
	Parse failures: no verdict / multiple verdicts	40/9
	Throughput on one H200 / mean output tokens	33.3 per s / 99.5
	First-shown verdicts: A / B / tie	6,943/2,345/2,663
Confirmation	Judge inputs truncated at 8192 tokens	0
	Individual-rubric cyclic triangles in panel A	24
	Cycle edges holding in both orders	15/72 = 0.208
	Eligible witness triangles	0 of 4 rubrics
	Judgments spent / confirmation budget	0/2,280

GPT: R29 — Complete the planned split change before using validation as evidence of generalization. The table explicitly distinguishes the current pair-level split from a planned prompt-level split. The reason is substantive: six unordered pairs can share the same four responses, so random pair splitting allows training and validation to contain overlapping prompts and candidates. Group by source prompt before generating or allocating pairs, keep swapped orders together, and remove cross-split duplicates. Explain which existing numbers use the old split and which use the corrected protocol; do not combine them as one held-out evaluation. The 657-prompt main panel and 698-prompt diagnostic panel also need explicit identities/overlap so apparent metric differences can be separated from changes in evaluation population.

Response: R29 — Partially addressed. The protocol groups normalized source prompts before splitting and keeps derived pairs and reversed orders together. Realized UF counts report 39,743 groups after near-duplicate merging, nine benchmark-overlap removals, and 38,240 eligible groups. Immutable split manifests, duplicate checks, and overlap of the historical 657- and 698-prompt panels still require artifact verification; historical and repaired panels remain separate.

The completed Nash/utilitarian comparison uses one outer stage and policy seeds 42/43/44. Earlier WBC and long-schedule diagnostics each use one policy seed. Comparisons within the repaired panel share the stated prompts, teacher, pools, and realization settings. Equal update counts do not imply equal FLOPs. Table 18 records the original recipe; resolved configurations and external-baseline results still require source verification.

E SOLVER AND NEURAL-REALIZATION DIAGNOSTICS

Completed SafeRLHF teacher-fit comparison. The completed study uses 2000 policy-training prompts, 500 development prompts, 1000 held-out prompts, and the fixed 250-update regression recipe. Both Nash and L1-matched utilitarian attain positive teacher-defined aggregate surplus across three seeds; their paired objective intervals include zero. The schedule and pool diagnostics below concern target realization and do not replace independent policy evaluation.

Table 18: Reported study dimensions and announced repair settings. Optimizer, precision, and sampling settings must be reconciled with the execution manifest. This table concerns the SafeRLHF panel; UltraFeedback settings appear in Appendix C.3.

Setting	Value
Base / adaptation	Llama-3.1-8B-Instruct / full fine-tuning
Train / dev / held-out prompts	2,000/500/1,000
Policy seeds / outer stages	42/43/44 (Nash/utilitarian); 1 stage
Learner / comparator occurrences	8/8; independent streams
Sampling	temperature 1, top- $p = 1$; preserve terminal tokens
Preference models	frozen three-seed objective ensembles
Inner solve / neural target	direct finite-pool / canonical log-ratio
Optimizer / learning rate	AdamW / 5×10^{-7}
Adam moments / epsilon	(0.9, 0.95) / 10^{-8}
Schedule / declared updates	cosine / 1750 or 250
Warmup ratio / weight decay	0.10/0
Global batch / pair rows	32 pairs / 56,000
Prompt / total sequence limit	1024/2048 tokens
Precision / gradient clip	bf16 forward, fp32 reduction / 1
Opponent temperatures / proximal η	— (resolved config required)
Reference cache / optimizer-state checks	— (execution report required)
Model / split / config revisions	— (immutable manifest required)

Table 19: Matched mechanism controls. External PROSPER and MOPO comparisons require their actual algorithms and separate tuning contracts; game-maxmin or WBC alone is not their reproduction.

method	within-objective representation	aggregation	changed component relative to NBPO
NBPO	adaptive regularized game	Nash	—
Fixed-reference Nash	fixed reference margin	Nash	removes adaptive opponent
BT-RM–Nash	scalar BT reward	Nash	imposes scalar reward representation
Game-utilitarian	adaptive regularized game	equal-weight sum	replaces Nash aggregation
Global game-maxmin	adaptive regularized game	global minimum	replaces Nash aggregation
BT-RM–utilitarian	scalar BT reward	equal-weight sum	scalar representation and aggregation control

GPT: R30 — Separate changing the training temperature from changing the evaluation ruler. The temperature study is only interpretable if every trained policy is also evaluated at a common fixed diagnostic temperature and under common independent metrics. Evaluating each policy with its own V_{k,β_k} changes both the optimizer and the quantity being compared. Show a small train-temperature by evaluation-temperature comparison, while retaining one prespecified common evaluation column for the main conclusion. Opponent entropy is useful but does not diagnose correctness: with only four candidates, concentration may reflect noisy margin estimates rather than meaningful adversarial pressure. Pair the temperature sweep with comparator-pool/ L scaling and exact-versus-estimated opponent error on a finite game, instead of reading lower entropy as evidence that robustness improved.

Response: R30 — Partially addressed. The evaluation contract now fixes the evaluator, comparator bank, and diagnostic temperatures within each panel regardless of training temperature. A controlled temperature study is described, but a train-temperature by evaluation-temperature matrix and exact-versus-estimated opponent errors across sampling and pool sizes are not documented as completed.

The direct solve concerns a finite pool at a declared tolerance. It does not make the preference model, sampling approximation, or neural fit exact. Solver consistency, held-out regression, and generated-policy evaluation are reported separately.

Alternative realization. For comparison with (26), weighted behavior cloning uses

$$\mathcal{L}_{\text{WBC}}^{(t)}(\theta) = \mathbb{E}_x \sum_{i=1}^{n_x} p_{t,x}^*(i) [-\log \pi_\theta(y_{x,i} | x)]. \quad (43)$$

Weights are detached and full-vocabulary response log-probabilities are summed over tokens. With IID reference-policy occurrences, this weighted likelihood approximates population forward-KL

Table 20: Completed SafeRLHF comparison, three policy seeds per method. Entries reproduce reported means and spreads; the spread convention and minimum/averaging order require artifact verification. Surplus uses the training teachers and does not certify population individual rationality.

Family	Seeds	nMSE↓	Sign↑	$\min_k s_k \uparrow$
NBPO-Nash	3	0.6845 ± 0.0048	0.7009 ± 0.0016	0.1215 ± 0.0028
Utilitarian (L1-matched)	3	0.6737 ± 0.0242	0.7040 ± 0.0058	0.1220 ± 0.0029

Table 21: Target realization on 500 development prompts, one seed. Each horizon has its own cosine schedule; 250-update runs are development-informed follow-ups. Pool KL is $D_{\text{KL}}(p^* \parallel \hat{p}_\theta)$ on sampled support; the reference value is 0.7879. Full diagnostics are in Appendix E.

Projection	Horizon	nMSE	Sign	Pearson	Pool KL↓	GPU-h
MSE	1750	1.418	0.693	0.415	1.2131	7.48
WBC	1750	13.070	0.486	-0.152	6.0634	6.78
MSE [†]	250	0.690	0.702	0.588	0.6204	2.20
WBC [†]	250	0.780	0.687	0.492	0.6875	2.08

projection; it remains a finite-sample realization rather than an exact population update (Agnihotri et al., 2026). Averaging $\frac{n_x}{2} [p_i^*(-\log \pi_\theta(y_i | x)) + p_j^*(-\log \pi_\theta(y_j | x))]$ over all unordered pairs equals the prompt-wise objective. Uniform teacher weights need not produce zero gradient on a finite sampled dataset.

Metrics and selection history. Write $t = \hat{h}_t^*$ and $h = h_t(\pi_\theta)$. We report $\text{nMSE} = \mathbb{E}[(h - t)^2] / \mathbb{E}[t^2]$, sign agreement, Pearson and Spearman correlations. Expectations use the stated split and pair measure. The zero predictor has nMSE 1. The reported oracle rescaling diagnostic is

$$a^* = \frac{\mathbb{E}[ht]}{\mathbb{E}[h^2]}, \quad \text{nMSE}^* = \min_{a \in \mathbb{R}} \frac{\mathbb{E}[(ah - t)^2]}{\mathbb{E}[t^2]}.$$

It fits a scalar on the evaluation observations and can reverse sign. It is neither centered explained variance nor a deployable calibration result. A direction-preserving version restricts $a \geq 0$; signed coefficients and this additional metric require the raw moments. If $\mathbb{E}[h^2] = 0$, set $a^* = 0$ and the residual to 1. Mean response log-ratio is measured under the sampled occurrence distribution; it is not total pool probability or a full-policy KL certificate.

GPT: R31 — Critical diagnostic: the reported regression fit may not beat predicting zero. The existing target RMS of 8.56 implies zero-output MSE $8.56^2 = 73.2736$, slightly below the reported fitted MSE 73.9, if both use identical examples, targets, and weighting. This would mean the learned log-ratio change is no better in this metric than making no update. First reconcile the measurement conventions and report initial/final loss against the same zero baseline. Then separate error against the conditional-mean target from irreducible binary-label variance using the frozen preference matrices, and run an oracle-target regression control. Vary target magnitude, neural optimization budget, and pair coverage. A very accurate frozen dual cannot compensate for failure to realize its target in the language model; resolve this before interpreting the main optimizer claim.

Response: R31 — Partially addressed. Inlined diagnostics now show UF-4 development nMSE 0.5861 ± 0.0033 across three policy seeds, below the zero-predictor baseline of 1, with sign accuracy and correlation reported separately. This establishes improved fitting to the frozen finite-pool target. The historical RMS/MSE conventions and a complete sampling-error versus neural-error decomposition remain unreconciled.

The 1750-update comparison was specified first. The 250-update schedules were chosen after inspecting development trajectories and declared before their runs. Each uses its own warmup and cosine horizon. A checkpoint at step 250 of the long run is therefore a different condition. Diagnostic thresholds were retained, but their use in development precludes describing the entire 2×2 as an untouched preregistered comparison.

Table 22: All recorded development evaluations of the repaired runs. The first column identifies the declared schedule; “Updates” is the checkpoint step. Oracle nMSE* permits signed rescaling. No trajectory point is substituted for its run’s declared final checkpoint.

Projection	Updates	nMSE	nMSE*	Sign acc.	Pearson	Spearman	$\log(\pi_\theta/\pi_t)$
NBPO-MSE, 1750	250	0.839	0.734	0.692	0.516	0.499	-1.13
	500	1.226	0.796	0.688	0.452	0.465	-1.86
	750	1.482	0.824	0.678	0.419	0.452	-2.42
	1000	1.568	0.839	0.694	0.401	0.449	-2.38
	1250	1.517	0.836	0.688	0.404	0.450	-2.27
	1500	1.432	0.831	0.691	0.412	0.455	-2.12
	1750	1.418	0.828	0.693	0.415	0.460	-2.07
NBPO-WBC, 1750	250	0.920	0.824	0.645	0.419	0.415	-0.75
	500	1.932	0.938	0.623	0.250	0.297	-2.04
	750	3.776	0.995	0.569	0.074	0.124	-3.27
	1000	6.121	0.999	0.535	-0.028	0.018	-4.39
	1250	11.611	0.979	0.496	-0.143	-0.096	-6.15
	1500	12.672	0.979	0.489	-0.144	-0.097	-6.63
	1750	13.070	0.977	0.486	-0.152	-0.105	-6.68
NBPO-MSE, 250 [†]	50	0.737	0.702	0.675	0.546	0.507	-0.77
	100	0.751	0.674	0.691	0.571	0.537	-1.11
	150	0.710	0.659	0.695	0.584	0.548	-1.03
	200	0.701	0.668	0.693	0.576	0.539	-0.87
	250	0.690	0.655	0.702	0.588	0.552	-0.90
NBPO-WBC, 250 [†]	50	0.866	0.839	0.628	0.401	0.388	-0.39
	100	0.807	0.777	0.655	0.473	0.450	-0.55
	150	0.755	0.730	0.678	0.520	0.499	-0.64
	200	0.770	0.743	0.682	0.507	0.497	-0.57
	250	0.780	0.758	0.687	0.492	0.488	-0.51

Table 23: Independent reconstruction on the development pool. Correlation intervals are prompt-clustered. KL improvement is reference minus policy KL. Target-argmax mass is $\hat{p}_\theta(i^*)$, $i^* \in \arg \max_i p^*(i)$; tie rules must be fixed. The WBC-1750 improvement is reported at source precision and differs by 10^{-4} from subtracting rounded KL entries.

Policy	Pearson [clustered CI]	$D_{\text{KL}} \downarrow$	KL improvement [†]	Top-1	Argmax mass
Base	—	0.7879	+0.0000	0.108	0.1250
MSE (1750)	+0.417 [0.370, 0.463]	1.2131	−0.4252	0.420	0.3135
WBC (1750)	−0.150 [−0.205, −0.095]	6.0634	−5.2754	0.124	0.1113
MSE (250) [†]	+0.589 [0.554, 0.622]	0.6204	+0.1675	0.406	0.2906
WBC (250) [†]	+0.496 [0.459, 0.531]	0.6875	+0.1004	0.362	0.2283

Finite-pool distribution fidelity. For a candidate occurrence i , write $\Delta_i = \log \pi_\theta(y_i | x) - \log \pi_t(y_i | x)$ and reconstruct

$$\hat{p}_\theta(i | x) = \frac{p_{t,x}(i)e^{\Delta_i}}{\sum_j p_{t,x}(j)e^{\Delta_j}}.$$

This self-normalized importance distribution is evaluated against the solved target by forward KL on the fixed sampled support. It does not measure the full LLM divergence or probability outside that support. MSE-250 and WBC-250 reduce KL by 0.1675 and 0.1004, respectively; the reported intervals in Table 23 concern Pearson correlation, not KL improvement. Paired prompt intervals for KL remain to be reported. With uniform occurrence weights and $\Delta = 0$, target-argmax mass is algebraically 1/8; reference top-1 agreement depends on tie-breaking.

Implementation history. The repaired study changed target construction, reference handling, tokenization, sampling, and training data jointly. Earlier failures cannot identify the effect of any single correction. Detailed debugging sweeps and the retired prompted-judge audit are retained in the code archive rather than used as performance evidence.

E.1 AGGREGATION-TARGET SENSITIVITY

On SafeRLHF, Nash weights (6.325, 6.186) give mean target TV 0.0008 relative to equal weights with the same ℓ_1 mass. Pairwise-target difference RMS is 0.0034, about 0.30% of target RMS 1.125.

Table 24: Measured direct-solver scaling by prompt count, pool size, and CPU workers. Times cover finite-pool solving and artifact generation, not LLM rollout or training.

prompts	pool	workers	solve (s)	dual evals	s/eval	proj. KKT	inner resid.	peak RSS (MB)	artifact (s)
200	4+4	32	1.5	31	0.047	3×10^{-14}	3×10^{-11}	467	0.01
200	8+8	32	1.7	35	0.049	4×10^{-15}	5×10^{-12}	498	0.02
200	16+16	32	1.8	32	0.055	2×10^{-14}	10^{-11}	499	0.02
1000	4+4	96	4.4	33	0.132	7×10^{-15}	2×10^{-11}	499	0.03
1000	8+8	96	3.2	26	0.122	4×10^{-15}	6×10^{-12}	504	0.04
1000	16+16	96	4.4	37	0.119	3×10^{-13}	10×10^{-11}	534	0.07
7000	4+4	96	10.6	28	0.379	7×10^{-10}	4×10^{-11}	534	0.11
7000	8+8	96	10.9	27	0.404	2×10^{-15}	3×10^{-11}	544	0.22
7000	16+16	96	15.3	30	0.508	10^{-14}	7×10^{-11}	633	0.39

On UF-4, mean TV is 0.0195 and relative target RMS is 3.44% (Table 25). Utilitarian attains higher unregularized F but incurs greater proximal KL, leaving Nash with higher $J = F - D/\eta$. Matching coefficient mass therefore does not match realized policy movement; target distance alone does not determine statistical power.

Table 25: Nash versus mass-matched utilitarian targets by panel. TV measures distribution distance; relative target RMS measures log-ratio differences. UF-4 quantities were reported as recomputed from stored arrays independently of the solver.

Diagnostic	Value
SafeRLHF-2k: mean target TV	0.0008
SafeRLHF-2k: maximum target TV	0.0075
SafeRLHF-2k: pairwise target difference RMS	0.0034
SafeRLHF-2k: target RMS on the aggregation panel	1.125
SafeRLHF-2k: Nash λ_{help}	6.325
SafeRLHF-2k: Nash λ_{safe}	6.186
UF-4: mean target TV	0.0195
UF-4: maximum target TV	0.1258
UF-4: pairwise target difference RMS	0.1857
UF-4: target RMS on the aggregation panel	5.4024
UF-4: relative pairwise target RMS	0.0344
UF-4: Nash λL_1	33.9235
UF-4: largest / smallest Nash λ	11.0426/7.3043
UF-4: Nash $F = \sum_k \log s_k$	-8.493723
UF-4: utilitarian F	-8.469298
UF-4: Nash realized $D(\pi \pi_t)$	1.080072
UF-4: utilitarian realized $D(\pi \pi_t)$	1.106423
UF-4: Nash $J = F - D/\eta$	-9.573795
UF-4: utilitarian J	-9.575722
UF-4: J gap, utilitarian minus Nash	-0.001927
UF-4: independent Nash dual residual on train	3.74×10^{-14}

F EVALUATION PROTOCOL

Common-prompt denominators for the UF-4 final evaluation. The planned final split is 2,000 prompts. A prompt enters the reported table only when every evaluated method has both presentation orders parsed on all four attributes, which leaves 1,763. Each method’s own valid count is larger, because the losses are not on the same prompts: 1,924, 1,916 and 1,935 for NBPO seeds 42/43/44 and 1,920 for the utilitarian control. Reporting those per-method counts side by side would compare arms on slightly different prompts, so the common set is used throughout and the difference is recorded here. Invalid judgments are left missing; none is recoded as a tie or a loss and none is selectively retried.

Table 26: Paired Nash-minus-utilitarian teacher-surplus differences on 1000 SafeRLHF prompts, conditional on each policy seed. Intervals retain the source-reported construction and all include zero; they do not establish equivalence.

Seed	Δs_{help}	Reported CI	Δs_{safe}	Reported CI
42	-0.0003	$[-0.0067, +0.0058]$	-0.0010	$[-0.0074, +0.0050]$
43	+0.0019	$[-0.0038, +0.0076]$	+0.0004	$[-0.0052, +0.0061]$
44	-0.0033	$[-0.0086, +0.0023]$	-0.0037	$[-0.0095, +0.0017]$

Table 27: Capability scores under each panel’s evaluation protocol. ARC-C/HellaSwag use normalized accuracy with 25/10 shots; MMLU uses 5-shot accuracy. UF IFEval is strict-prompt accuracy; SafeRLHF means and sample standard deviations use three seeds, with its IFEval variant still requiring verification.

Method	ARC-C	HellaSwag	MMLU	IFEval	GSM8K	HB↓
<i>UltraFeedback: one policy seed</i>						
Base	0.6049	0.8008	0.6823	0.7542	0.8666	0.2813
NBPO	0.6058	0.7984	0.6816	0.7227	0.8605	—
Game-utilitarian	0.6032	0.7978	0.6817	0.7209	0.8658	—
<i>SafeRLHF: reported three-seed study</i>						
Base	—	—	—	0.752	0.867	0.281
NBPO	—	—	—	0.748 ± 0.013	—	—
Game-utilitarian	—	—	—	0.745 ± 0.025	—	—

F.1 COMPLETE CAPABILITY RESULTS

This table retains the full reported capability panel. The UF and historical SafeRLHF base rows use different recorded protocols and are not pooled. The UF rows are policy seed 42. ARC-C, HellaSwag and MMLU are all complete on all three policy seeds of both families. Seed means with sample standard deviations are 0.6064 ± 0.0010 , 0.7994 ± 0.0011 and 0.6820 ± 0.0006 for NBPO, and 0.6044 ± 0.0013 , 0.7991 ± 0.0012 and 0.6822 ± 0.0005 for game-utilitarian, against base values of 0.6049, 0.8008 and 0.6823. Every seed-mean change from the base is within 0.0017 (largest: game-utilitarian HellaSwag, -0.0017), no individual cell differs by more than 0.0030, and the seed spreads are 0.0005–0.0013. These are point estimates on fixed benchmarks, not equivalence tests. Table 3 in the main text isolates the core outcomes to complete on the UF checkpoints.

Teacher-defined and independent evaluation. UltraFeedback policy evaluation uses the frozen 2000-prompt final split and the same cached base response for both arms. A separate local judge evaluates four rubrics in both presentation orders, with ties worth 0.5, for 16,000 scheduled judgments per arm. Retained counts are 1967 for NBPO and 1960 for utilitarian. The source reports 2000 whole-prompt bootstrap replicates on roughly 1950 common prompts and no resolved paired advantage; numerical UF intervals and the policy-evaluation judge revision are not supplied. These results condition on one policy seed.

GPT: R33 — Make label provenance and rubric identity auditable for every reported run. The note distinguishing Beaver-head precursor runs from prompted comparisons is important. Scalar-head-induced rankings are transitive by construction and therefore cannot substantiate the paper’s claim about preserving within-objective cycles. Attach an unambiguous label-source identifier to each checkpoint/result and ensure that the main adaptive versus fixed-reference comparison differs only in the intended component. Also provide the actual per-panel rubric mapping: the prose lists correctness/coherence, whereas some tables use honesty, truthfulness, and instruction following. This is not just terminology because changing or splitting an objective changes both the preference matrices and the bargaining outcome. Measure rubric agreement and show independent evaluator results separately before pooling them.

Response: R33 — Partially addressed. The main text identifies score-induced UF supervision, human-pairwise SafeRLHF, controlled games, and the separate direct-judgment audit by their distinct roles. The cross-play template requires a common independent evaluator and frozen comparator bank. Full checkpoint provenance, evaluator revisions, and rubric-agreement measurements still require execution artifacts.

Table 28: Historical single-checkpoint objective evaluations. Short-run objective entries retain their reported precision. These rows do not replace the completed three-seed comparison in Table 20.

Method	Updates	$s_{\text{help}} \uparrow$	$s_{\text{safe}} \uparrow$	$\min_k s_k \uparrow$
Finite-pool target p^*	—	+0.1653	+0.1662	+0.1653
Base (greedy)	—	−0.0405	−0.0320	−0.0405
NBPO-MSE (1750)	1750	+0.1339	+0.1337	+0.1337
NBPO-WBC (1750)	1750	+0.0738	+0.0808	+0.0738
NBPO-MSE (250) [†]	250	+0.119	+0.125	+0.1186
NBPO-WBC (250) [†]	250	—	—	+0.0569
Game-utilitarian	250	—	—	—
Fixed-reference Nash	250	—	—	—
BT-RM-Nash	250	—	—	—
PROSPER	—	—	—	—
MOPO	—	—	—	—
Game-maxmin	250	—	—	—
Scalarized DPO	—	—	—	—

Table 29: Completed downstream snapshot supplied by the execution report. Arena is an unassigned local-proxy aggregate; token medians have an unassigned prompt scope. These values are preserved without relabeling them as hard/creative or Alpaca-specific results. This historical snapshot contains one checkpoint per row; it is not the new three-seed aggregate.

Policy	$\min_k s_k$	IFEval	GSM8K	Harm↓	Alpaca	Arena (unassigned)	Tokens (unassigned)
Base	−0.0405	.752	.867	.281	.500	.500	457
MSE (1750)	+0.1337	.758	.864	.284	.419	.435	481
WBC (1750)	+0.0738	.741	.852	.256	.578	.594	457
MSE (250) [†]	+0.1186	.760	.870	.269	.497	.516	460
WBC (250) [†]	+0.0569	.741	.867	.266	.486	.542	463

SafeRLHF surplus uses the frozen training preference models on held-out prompts, common comparator pools, and the same definition of $\hat{d}$. Greedy responses define point-mass decoders; they are not unbiased samples of the trained stochastic policy. Their positive surplus does not certify Algorithm 1 acceptance. Any stochastic-policy evaluation must sample both the trained policy and reference under a common declared protocol and account for prompt uncertainty.

IFEval, GSM8K, harmful-completion checks, and benign-refusal evaluation measure complementary behavior. AlpacaEval and Arena-Hard prompt sets are scored by a local reward model; these are not official leaderboard scores. Table 29 preserves the execution report’s Arena aggregate and token medians, whose subset and prompt scope remain unspecified. They do not populate subset-specific main-table cells. The exact IFEval strict metric and numerical confidence intervals must be reconciled with evaluator outputs. A confidence interval containing 0.5 does not establish win-rate equivalence.

Refusal concentration and teacher truncation. The audit distinguishes identical-string responses (247/1000), short responses (450/1000), refusal-prefixed responses (621/1000), and refusals among short responses (448/450). These output-defined categories overlap and are not interchangeable. Gains in a refusal subgroup do not identify a Nash-specific mechanism.

Teacher truncation affects $26,790/84,000 = 31.89\%$ of candidate occurrences; prompt truncation affects one occurrence. Among truncated responses, the reported median loss is 232 tokens and the observed response fraction is 61%. The latter quantities must retain their aggregation definitions. This is a response-tail visibility issue for the implemented teacher. The UltraFeedback panel therefore uses a natively supported longer-context encoder and a separate full-response evaluator, rather than increasing RoBERTa’s length limit. Reported joint-positive rates 0.725/0.730 are archived with their unit unresolved; they are not population IR certificates.

Table 30: Evaluation contracts and fields to complete from actual artifacts. These counts specify the requested full evaluation sets, not a claim that every job has completed.

Set / target count	Metric and completion contract
SafeRLHF / 1000	Teacher-defined objective values; comparator and disagreement hashes; greedy and stochastic evaluations kept separate.
IFEval / 541	Official strict prompt and instruction accuracy; loose scores secondary. Verify the reported strict column against evaluator output.
GSM8K / 1319	Fixed zero-shot-CoT prompt and answer extractor; exact match and parse-failure count.
HarmBench / text test	Report actual category counts and classifier version. Use category-specific hashing where required.
XSTest / 250 safe, 200 unsafe	Benign refusal and unsafe-request handling separately; report classifier or blinded manual assessment.
Alpaca / 805	Same-prompt local reward-model win rate versus base, ties half; response length from the same prompt set.
Arena / 500 hard, 250 creative	Report subset-specific local proxy win rates; an unassigned aggregate does not populate these columns.
MT-Bench / 80	Optional two-turn responses; no official score without the stated judging protocol.

F.2 INDEPENDENT CROSS-PLAY AND FINITE-BANK DIAGNOSTICS

Within each panel, fix the evaluator, rubric wording, parsing, response sampling, and held-out prompt IDs before comparisons. The comparator bank contains the base, NBPO, fixed-reference Nash, and one strong competitor selected on development data. Use fresh responses outside the training pool and both presentation orders. Comparator selection does not access final outcomes. Table 31 records each rubric separately; all cells remain unmeasured.

For a generated policy pair, let $\hat{J}_k(\pi, q) = \widehat{W}_k(\pi, q) - 1/2$. A complementary policy-bank diagnostic is

$$\widehat{V}_k^{\mathcal{B}}(\pi) = \min_{w \in \Delta(\mathcal{B})} \left\{ \sum_{q \in \mathcal{B}} w_q \hat{J}_k(\pi, q) + \beta_k \text{KL}(w \| a) \right\}, \quad \widehat{s}_k^{\mathcal{B}}(\pi) = \widehat{V}_k^{\mathcal{B}}(\pi) - \widehat{V}_k^{\mathcal{B}}(\pi_{\text{ref}}).$$

Here a is a full-support distribution over the comparator policies, fixed on development data and shared by every evaluated policy; β_k is also fixed within the panel. Record both before evaluating this diagnostic. This is a regularized finite-policy-bank game. Its KL is over comparator weights and is not the response-distribution KL in (6); consequently it is not a direct estimate of the population value V_k or population exploitability. Training-teacher values are reported separately.

Table 31: Independent cross-play template: row-policy win rates against column comparators, separately for each UF rubric. The competitor is selected on development data. Dashes are unmeasured.

Rubric	Policy	Base	NBPO	Fixed-ref.	Competitor
IF	Base	—	—	—	—
	NBPO	—	—	—	—
	Fixed-reference Nash	—	—	—	—
	Competitor	—	—	—	—
Truth	Base	—	—	—	—
	NBPO	—	—	—	—
	Fixed-reference Nash	—	—	—	—
	Competitor	—	—	—	—
Honesty	Base	—	—	—	—
	NBPO	—	—	—	—
	Fixed-reference Nash	—	—	—	—
	Competitor	—	—	—	—
Help	Base	—	—	—	—
	NBPO	—	—	—	—
	Fixed-reference Nash	—	—	—	—
	Competitor	—	—	—	—

Use a declared common valid subset for the cross-play summary and report coverage by method, rubric, and presentation order. Invalid judgments remain missing. Whole-prompt paired resampling keeps the policy pairs and rubrics together, recomputes each minimum and the finite-bank solve, and reports numerical intervals. Training-seed variation is reported separately. The historical 1967/1960

UF counts are not an exact paired denominator; recover the common IDs from the original judgments before completing that comparison. Any new analysis choices retain the history of already observed final results.

F.3 FEEDBACK BUDGETS AND PROVENANCE

We record unique semantic comparisons separately from order-swapped calls, generated tokens, preference-model fitting/scoring, solver time, neural training, and tuning trials. Numerical iterations reuse cached matrices and do not supply new labels. Budget-matched claims require a shared feedback contract and total-compute accounting. All responses, derived pairs, and reversed orders from one source prompt remain in the same split. Manifests identify feedback source, teacher/rubric revisions, objective order, split, comparator construction, and decoding. Historical pair-split, precursor, or differently judged results are not pooled with the repaired panel. Instruction-following error types, length, and refusal patterns are descriptive diagnostics; output-defined subgroups alone do not identify a causal mechanism.

Table 32: Cost-accounting template. Teacher, policy, generation/evaluation, and total costs use GPU-hours; total includes declared tuning. Solver CPU time is reported separately. Policy cost is per policy seed and is the trainer-reported runtime times the four devices held. On this measure the seven completed UF arms agree to within 0.02 GPU-hours (11.17–11.19), so the fixed-reference and utilitarian entries are the same 11.2 as NBPO rather than an independent coincidence; three policy seeds are trained for NBPO and for game-utilitarian, one for fixed-reference Nash. The controller-side wall cost, which also covers model loading and checkpointing, is 12.25 GPU-hours per seed, and NBPO seed 42 additionally consumed 10.1 GPU-hours in two failed attempts before the run that is reported. Remaining columns are unmeasured; unique labels in particular await the declared convention separating unique semantic comparisons from order-swapped calls.

Method	Unique labels	Teacher	Policy	Gen./eval.	Total	Trials
NBPO, seed 42	—	—	11.2	—	—	—
Fixed-reference Nash	—	—	11.2	—	—	—
Game-utilitarian	—	—	11.2	—	—	—
BT-RM–Nash	—	—	—	—	—	—
Global game-maxmin	—	—	—	—	—	—
PROSPER (adapt.)	—	—	—	—	—	—
MOPO (adapt.)	—	—	—	—	—	—
DPO (uniform / sweep)	—	—	—	—	—	—

Report generated/scored token counts, order-swapped and repeated judgments, solver CPU time, and shared-cache attribution alongside Table 32. The DPO deployment-policy cost and total seven-weight search cost are distinct. Existing component measurements do not establish an equal-total-budget comparison or a performance-versus-compute curve.

F.4 EMPIRICAL PARETO EVALUATION

The population Nash optimum is Pareto efficient under the stated assumptions; a single proximal step and approximate neural realization do not inherit this certificate. The instruction comparison evaluates all four attributes using a fixed independent local judge; current policy results cover NBPO and the utilitarian control at one seed. A policy is empirically nondominated only relative to the evaluated policies and evaluator in this four-dimensional space. The two projections in Figure 1 cannot establish four-dimensional dominance.

Scalarized DPO (Rafailov et al., 2023) uses scalar BT rewards calibrated on development data. The declared weight set contains the four coordinate vectors, $(.25, .25, .25, .25)$, $(.1, .1, .1, .7)$, and $(.1, .7, .1, .1)$, in IF/TF/Hon/Help order. Its main-table row denotes the uniform setting; the full weight sweep remains unmeasured. Actual MOPO and PROSPER adaptations preserve their respective constraints and game updates. The DPO sweep and external-method rows remain unmeasured.

Compare methods on the complete final set and prespecified label-disagreement subsets. Keep independent win rates separate from training-teacher game values. Report individual policy seeds, paired prompt uncertainty, and all declared weights. Curves joining points do not represent evaluated

mixture policies. Nash welfare is defined only for positive game surpluses; mark other cases infeasible. Fix hypervolume normalization and reference points before final scores are opened.

Uncertainty and provenance. Report every policy seed, then mean and standard deviation across training seeds. Each paired prompt-bootstrap replicate recomputes objective means and then the minimum. Its intervals condition on a checkpoint and do not include training-seed variability. Freeze development choices before confirmatory comparisons and retain the history of benchmark exposure. SafeRLHF Nash/utilitarian aggregates and UF neural diagnostics use three policy seeds; UF generated-policy comparisons and historical WBC/long-schedule snapshots use one. Missing method-wise values must not be reconstructed from ranges pooled across methods; dashes are unmeasured, not zeros. All result updates must record checkpoint, split, decoding, evaluator, sample count, and source artifact. Unmatched literature scores do not fill matched-baseline rows.